Floer Homology

Lectures by Urs Frauenfelder

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1 Introduction

Hamiltonian Floer homology was introduced by Andreas Floer in the 80s to tackle the *homological Arnold conjecture*. Roughly speaking, the conjecture says in its simplest form that the number of nondegenerate solutions of a 1-periodic Hamiltonian equation is bounded below by the sum of the dimension of the singular homology of the compact symplectic manifold with coefficients in $\mathbb{Z}_2$. This resembles the famous *Morse inequalities*, and thus it is not surprising that Floer homology was largely influenced by Morse homology. However, a key technical ingredient in the construction in this semi-infinite dimensional version of Morse homology was Gromov's analysis of pseudoholomorhic curves. However, a key technical ingredient in the construction of this semi-infinite dimensional version of Morse homology was Gromov's analysis of pseudoholomorhic curves. Floer homology is constructed as the Morse homology of the *symplectic action functional*, which is a functional defined on the contractible component of the free loop space on a symplectic manifold, having precisely the 1-periodic orbits of the Hamiltonian vector field as critical points. A delicate analysis which uses both a geometrical assumption on the symplectic manifold as well as technical results originating in harmonic analysis, one shows that this Morse homology is indeed well-defined. In contrast to standard finite-dimensional Morse homology, new phenomenons occur for sequences of negative gradient flow lines, namely *bubbling* does occur. This theory is well understood in the case of Hamiltonian Floer homology of a symplectically aspherical symplectic manifold and is not original. However, some of the exposition cannot be found written out in the literature and we try to provide a comprehensive but also straight forward proof of the compactness in Hamiltonian Floer homology. This manuscript is a transcribed version of the lectures given by Prof. Dr. Urs Frauenfelder at the University of Augsburg in the winter term 2021. There are a lot of brilliant notes and textbooks on Floer homology, see for example [Sal99], [MS12] or [AD14].

2 The Symplectic Action Functional and Floer's Equation

2.1 The Symplectic Action Functional

A symplectic manifold is a pair (M, ω), where M is a finite-dimensional smooth manifold manifold without boundary of class C^∞, and $\omega \in \Omega^2(M)$ is closed and nondegenerate. Explicitly, the map

$$TM \to T^*M, \qquad (q, v) \mapsto i_v \omega_q,$$

is a smooth bundle isomorphism, called the *tangent-cotangent bundle isomorphism*. In these notes we will not study arbitrary compact symplectic manifolds, but rather special ones.

Definition 2.1 (Symplectic Asphericity). *A connected symplectic manifold (M, ω) is said to be* ***symplectically aspherical****, iff*

$$\int_{\mathbb{S}^2} f^*\omega = 0 \qquad \forall\, f \in C^\infty(\mathbb{S}^2, M).$$

Example 2.1 (Exact Symplectic Manifold). Every connected exact symplectic manifold (M, λ) is symplectically aspherical. Indeed, we compute

$$\int_{\mathbb{S}^2} f^* d\lambda = \int_{\mathbb{S}^2} df^*\lambda = \int_{\partial \mathbb{S}^2} f^*\lambda = 0$$

for all $f \in C^\infty(\mathbb{S}^2, M)$ by Stoke's theorem.

Example 2.2 ($\boldsymbol{\pi_2(M) = 0}$). Let (M, ω) be a connected symplectic manifold. As ω is closed, we have a well-defined homomorphism

$$\pi_2(M) \to \mathbb{R}, \qquad [f] \mapsto \int_{\mathbb{S}^2} f^*\omega.$$

Indeed, let $F \in C^\infty(\mathbb{S}^2 \times [0,1], M)$ be a homotopy between $f_0 \in C^\infty(\mathbb{S}^2, M)$ and $f_1 \in C^\infty(\mathbb{S}^2, M)$, then we compute

$$0 = \int_{\mathbb{S}^2 \times [0,1]} F^* d\omega = \int_{\mathbb{S}^2 \times [0,1]} dF^*\omega = \int_{\mathbb{S}^2} f_1^*\omega - \int_{\mathbb{S}^2} f_0^*\omega$$

by Stoke's theorem.

We denote by $\mathscr{L}M = C^\infty(\mathbb{S}^1, M)$ the free loop space, and $\mathscr{L}_0 M \subseteq \mathscr{L}M$ the connected component of contractible loops.

Definition 2.2 (The Symplectic Action Functional). *Let (M, ω) be a symplectically aspherical manifold. For $H \in C^\infty(M \times \mathbb{S}^1)$, define the* ***symplectic action functional***

$$\mathscr{A}_H \colon \mathscr{L}_0 M \to \mathbb{R}, \qquad \mathscr{A}_H(\gamma) := -\int_{\mathbb{D}} \bar{\gamma}^*\omega + \int_0^1 H_t(\gamma(t))dt, \tag{1}$$

where $\bar{\gamma} \in C^\infty(\mathbb{D}, M)$ is a filling of γ, that is, $\bar{\gamma}(e^{2\pi i t}) = \gamma(t)$ for all $t \in \mathbb{S}^1$.

Remark 2.1. As (M,ω) is symplectically aspherical, the symplectic action functional (1) is well-defined. Indeed, if $\bar{\gamma}_1, \bar{\gamma}_2 \in C^\infty(\mathbb{D}, M)$ are two filling discs of $\gamma \in \mathscr{L}_0 M$, then we obtain $f \in C^\infty(\mathbb{S}^2, M)$ by gluing the two discs together. But then

$$0 = \int_{\mathbb{S}^2} f^*\omega = \int_{\mathbb{D}} \bar{\gamma}_2^*\omega - \int_{\mathbb{D}} \bar{\gamma}_1^*\omega.$$

See Figure 1.

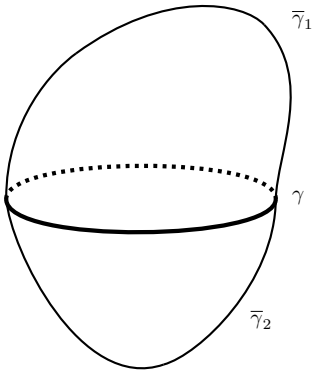


Figure 1: Two different filling discs $\bar{\gamma}_1, \bar{\gamma}_2 \in C^\infty(\mathbb{D}, M)$ of $\gamma \in \mathscr{L}_0 M$ give rise to a sphere.

Definition 2.3 (Differential of the Symplectic Action Functional). *Let (M,ω) be a symplectically aspherical manifold. For $H \in C^\infty(M \times \mathbb{S}^1)$ define the* ***differential of the symplectic action functional $\mathscr{A}_H$*** *by*

$$d\mathscr{A}_H|_\gamma \colon \Gamma(\gamma^* TM) \to \mathbb{R}, \qquad d\mathscr{A}_H|_\gamma(X) := \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} \mathscr{A}_H(\gamma_\varepsilon),$$

for all $\gamma \in \mathscr{L}_0 M$, where γ_ε is the exponential variation with respect to some Riemannian metric on M given by

$$\gamma_\varepsilon(t) := \exp_{\gamma(t)}(\varepsilon X(t)) \qquad \forall (t,\varepsilon) \in \mathbb{S}^1 \times (-\varepsilon_0, \varepsilon_0)\,,$$

for some $\varepsilon_0 > 0$ sufficiently small.

Definition 2.4 (Periodic Orbits). *Let $H \in C^\infty(M \times \mathbb{S}^1)$ for a symplectic manifold (M,ω). Define the set of* ***1-periodic contractible orbits of X_H*** *by*

$$\mathscr{P}_0(H) := \{\gamma \in \mathscr{L}_0 M : \dot{\gamma}(t) = X_{H_t}(\gamma(t))\ \forall t \in \mathbb{S}^1\}.$$

In fact, the set $\mathscr{P}_0(H)$ is a finite discrete set due to nondegeneracy of H and compactness of M.

Lemma 2.1 (Critical Points of the Symplectic Action Functional). *Let (M,ω) be a symplectically aspherical manifold and $H \in C^\infty(M \times \mathbb{S}^1)$. Then*

$$\operatorname{Crit} \mathscr{A}_H = \mathscr{P}_0(H).$$

Proof. For $\gamma_\varepsilon \in \mathscr{L}_0 M$ and $\xi \in \Gamma(\gamma^* TM)$, we choose capping $\bar{\gamma}_\varepsilon \in C^\infty(\mathbb{D}, M)$, and define $\bar{\xi} := \frac{\partial}{\partial \varepsilon}\big|_{\varepsilon=0} \bar{\gamma}_\varepsilon \in \Gamma(\bar{\gamma}^* TM)$. The variation of $\mathscr{A}_H$ along γ is computed as

$$\begin{aligned}
d\mathscr{A}_H|_\gamma(\xi) &= \frac{\partial}{\partial \varepsilon}\bigg|_{\varepsilon=0} \mathscr{A}_H(\gamma_\varepsilon) = \frac{\partial}{\partial \varepsilon}\bigg|_{\varepsilon=0} \left(-\int_{\mathbb{D}} \bar{\gamma}_\varepsilon^* \omega + \int_0^1 H_t(\gamma_\varepsilon(t))dt \right) \\
&= -\int_{\mathbb{D}} \bar{\gamma}^* L_{\bar{\xi}} \omega + \int_0^1 (dH_t)_{\gamma(t)}(\xi(t))dt \\
&\overset{\text{Cartan's formula}}{=} -\int_{\mathbb{D}} \bar{\gamma}^* i_{\bar{\xi}} d\omega - \int_{\mathbb{D}} d\bar{\gamma}^* i_{\bar{\xi}} \omega + \int_0^1 (dH_t)_{\gamma(t)}(\xi(t))dt \\
&\overset{\text{Stokes' thm.}}{=} -\int_{\mathbb{S}^1} \gamma^* i_\xi \omega + \int_0^1 (dH_t)_{\gamma(t)}(\xi(t))dt \\
&= -\int_0^1 \omega(\xi(t), \dot{\gamma}(t))dt + \int_0^1 \omega(\xi(t), X_{H_t}(\gamma(t)))dt \\
&= \int_0^1 \omega(\dot{\gamma}(t) - X_{H_t}(\gamma(t)), \xi(t))dt. \qquad (2)
\end{aligned}$$

The statement follows from the nondegeneracy of ω. □

We introduce a Morse condition for the symplectic action functional.

Definition 2.5 (Nondegenerate Critical Point). *Let (M,ω) be a symplectically aspherical manifold and $H \in C^\infty(M \times \mathbb{S}^1)$. A critical point $\gamma \in \operatorname{Crit} \mathscr{A}_H$ is said to be **nondegenerate**, iff*

$$D\phi_1^H|_{\gamma(0)} - \mathrm{id}_{T_{\gamma(0)}M}$$

is invertible, where ϕ^H denotes the flow of the time-dependent Hamiltonian vector field X_H. A Hamiltonian function $H \in C^\infty(M \times \mathbb{S}^1)$ is said to be non-degenerate, iff every $\gamma \in \mathscr{P}_0(H)$ is non-degenerate.

Remark 2.2. Let $H \in C^\infty(M)$ for a symplectic manifold (M,ω). If $x \in \operatorname{Crit} H$ and the constant loop $c_x \in \mathscr{L}_0 M$ is non-degenerate, then $\ker \operatorname{Hess} H(x)$ is trivial. If $x \notin \operatorname{Crit} H$, then every $\gamma \in \mathscr{P}_0(H)$ is degenerate. Indeed, we have that

$$D\phi_1^H(X_H(x)) = X_H(x).$$

The nondegeneracy condition therefore also reads:

$$\det(D\phi_1^H|_x - \mathrm{id}_{T_x M}) \neq 0 \qquad \forall\, x \in M.$$

Remark 2.3. That nondegeneracy as stated above is indeed a Morse condition for the symplectic action functional $\mathscr{A}_H$ can be seen as follows. For every $\gamma \in \operatorname{Crit} \mathscr{A}_H$ there is a canonical isomorphism:

$$\ker \operatorname{Hess} \mathscr{A}_H|_\gamma \cong \{X \in T_{\gamma(0)}M : D\phi_1^H(X) = X\},$$

and the right hand side is $\{0\}$ by definition of H being non-degenerate.

Now we can formulate the main theorem in this note.

Theorem 2.1 (Homological Arnold Conjecture in the Symplectically Aspherical Case). *Let $H \in C^\infty(M \times \mathbb{S}^1)$ be a nondegenerate Hamiltonian function on a closed symplectically aspherical manifold (M, ω). Then number of 1-periodic orbits of the Hamiltonian vector field X_H is bounded below by the sum of Betti numbers, that is,*

$$\#\mathscr{P}_0(H) \geq \sum_{k=0}^{\dim M} \dim_{\mathbb{Z}_2} \mathrm{H}_k(M; \mathbb{Z}_2). \tag{3}$$

Remark 2.4. A 1-periodic orbit γ can be equivalently regarded as a fixed point $x = \gamma(0)$ of the Hamiltonian symplectomorphism $\varphi_H := \phi_1^H \in \operatorname{Ham}(M, \omega)$, i.e. $\operatorname{Fix}(\varphi_H) = \mathscr{P}_0(H)$. Thus, the Arnold conjecture above can also be rephrased as: For any nondegenerate Hamiltonian diffeomorphism φ_H on a closed symplectically aspherical manifold (M, ω),

$$\#\operatorname{Fix}(\varphi_H) \geq \sum_{k=0}^{\dim M} \dim_{\mathbb{Z}_2} \mathrm{H}_k(M; \mathbb{Z}_2).$$

The statement of homological Arnold conjecture is analogous to the Morse inequality in the finite dimensional Morse theory, which can be proved via the Morse homology for the Morse function, c.f. [FN20, Theorem 3.1]. If we could associate some proper notion of Morse homology for the symplectic action functional $\mathscr{A}_H : \mathscr{L}_0 M \to \mathbb{R}$, we could also treat the Arnold conjecture as certain guise of Morse inequality. This is originally the main purpose of Hamiltonian Floer theory, and we will see, after establishing the Floer homology, that it is not quite the Morse homology for the infinite-dimensional non-compact smooth manifold $\mathscr{L}_0 M$, which is one reason why it is regarded only as semi-infinite dimensional version Morse homology.

2.2 Negative Gradient Flow Equation and Floer Equation

Definition 2.6 (L^2-inner product). *Let (M, ω) be a symplectic manifold and $H \in C^\infty(M \times \mathbb{S}^1)$. For a smooth family $(J_t)_{t \in \mathbb{S}^1}$ of ω-compatible almost complex structures on (M, ω), the **L^2-inner product on $\mathscr{L}M$** is defined as*

$$\langle \xi, \eta \rangle_J := \int_0^1 g_{J_t}(\xi(t), \eta(t))\, dt := \int_0^1 \omega(\xi(t), J_t \eta(t))\, dt \tag{4}$$

for all $\xi, \eta \in \Gamma(\gamma^* TM)$, and $\gamma \in \mathscr{L}M$.

Remark 2.5. The L^2-inner product (4) makes $\mathscr{L}M$ become a Hilbert manifold.

On symplectically aspherical manifold (M,ω), the formal gradient $\operatorname{grad}_J \mathscr{A}_H \in \Gamma(T\mathscr{L}_0 M)$ of the symplectic action functional $\mathscr{A}_H$ with respect to the L^2-inner product (4) defined by

$$\langle \operatorname{grad}_J \mathscr{A}_H, \cdot \rangle_J = d\mathscr{A}_H$$

is given by

$$\operatorname{grad}_J \mathscr{A}_H(\gamma(t)) = J_t(\gamma(t))(\dot{\gamma}(t) - X_{H_t}(\gamma(t))) \qquad \forall\, t \in \mathbb{S}^1. \tag{5}$$

Indeed, this readily follows from (2). Note that $\operatorname{grad}_J \mathscr{A}_H$ is not necessarily tangent to $\mathscr{L}_0 M$, since we only know that it lies in the L^2-completion of $\Gamma(T\mathscr{L}_0 M)$.

Definition 2.7 (Negative Gradient Flow Line). *Let (M,ω) be a symplectically aspherical manifold and $H \in C^\infty(M \times \mathbb{S}^1)$. A* ***negative gradient flow line*** *of the symplectic action functional $\mathscr{A}_H$ is formally defined to be a map $u \in C^\infty(\mathbb{R}, \mathscr{L}_0 M)$ such that*

$$\dot{u}(s) = -\operatorname{grad}_J \mathscr{A}_H(u(s)) \qquad \forall\, s \in \mathbb{R}. \tag{6}$$

Remark 2.6. The symplectic action functional $\mathscr{A}_H$ is a Lyapunov function for the negative gradient flow of $\mathscr{A}_H$. Indeed, we compute

$$\begin{aligned} \frac{d}{ds}\mathscr{A}_H(u(s)) &= d\mathscr{A}_H(\dot{u}(s)) \\ &= \langle \operatorname{grad}_J \mathscr{A}_H(u(s)), \dot{u}(s) \rangle_J \\ &= -\langle \operatorname{grad}_J \mathscr{A}_H(u(s)), \operatorname{grad}_J \mathscr{A}_H(u(s)) \rangle_J \\ &= -\left\| \operatorname{grad}_J \mathscr{A}_H(u(s)) \right\|_J^2 \end{aligned}$$

for all $s \in \mathbb{R}$ and negative gradient flow lines u. Note that such flow is not really a regularized negative gradient flow of the Morse function $\mathscr{A}_H$ on the free loop space $\mathscr{L}_0 M$ comparing to the finite dimensional case, as it is not true that for any given initial value $u(0) = \gamma \in \mathscr{L}_0 M$ the equation (6) has a solution for such Cauchy initial value problem. Instead, we will see later that such unregularized flow is solving asymptotic boundary condition problem by regarding (6) as a non-linear first order elliptic PDE.

A negative gradient flow line u of the symplectic action functional $\mathscr{A}_H$ is formally a smooth map in $C^\infty(\mathbb{R}, \mathscr{L}_0 M)$. As $\mathscr{L}_0 M$ is only a Fréchet manifold, studying the negative gradient flow equation (6) as an ordinary differential equation on $\mathscr{L}_0 M$ is difficult and problematic. Floer's ingenious idea was to interpret the negative gradient flow equation as a partial differential equation on M, that is, u is interpreted as a smooth function on the cylinder

$$\mathbb{R} \times \mathbb{S}^1 \to M, \qquad (s,t) \mapsto u(s)(t).$$

Explicitly, a negative gradient flow line of the symplectic action functional $\mathscr{A}_H$ is a map $u \in C^\infty(\mathbb{R} \times \mathbb{S}^1, M)$ solving the partial differential equation

$$\partial_s u(s,t) + J_t(u(s,t))(\partial_t u(s,t) - X_{H_t}(u(s,t))) = 0 \qquad \forall\, (s,t) \in \mathbb{R} \times \mathbb{S}^1. \tag{7}$$

Equation (7) is called ***Floer's equation***, and the solutions are called the ***Floer cylinders***. When $H_t = 0$, the Floer equation (7) becomes a nonlinear Cauchy–Riemann equation, and if further $J_t = J$ then u is a J-holomorphic cylinder. See [MS12, p. 19]. When $H_t \neq 0$, as there are not any derivatives of u w.r.t. variables s, t in the term $X_{H_t}(u)$, such term is regarded as zero order perturbed term on nonlinear Cauchy–Riemann operator. Thus the Floer equation is a perturbed nonlinear Cauchy–Riemann equation, and therefore a first order nonlinear elliptic partial differential equation.

3 Compactness

In order to build a Morse homology associated with a Morse function, compactness of the moduli space of negative gradient flow lines is crucial. On symplectic aspherical manifold (M,ω), the action functional $\mathscr{A}_H : \mathscr{L}_0 M \to \mathbb{R}$ for nondegenerate Hamiltonian H_t is regarded as a Morse function. The difference is that now $\mathscr{L}_0 M$ is an infinite dimensional non-compact smooth manifold. Nevertheless, we could still obtain similar compactness result for the moduli space of negative gradient flow lines, i.e. the Floer cylinders, in this infinite dimensional scenario. We introduce proper notions, e.g. broken Floer cylinders and Floer-Gromov convergence, as in the finite-dimensional Morse homology in [FN20], and state the Floer-Gromov compactness for Floer cylinders. We put main effort to discuss the essential distinction from finite dimensional case – bubbling phenomenon, which would be ruled out thanks to the symplectic asphericity. The upshot is that it is only under the crucial assumption of symplectic asphericity could we obtain the analogous compactness pattern for Floer cylinders as in finite dimensional Morse homology.

3.1 Floer-Gromov Convergence

In the finite dimensional Morse theory, we have the asymptotic convergence of finite energy negative gradient flow lines towards critical points of the Morse function. See [FN20, §4.3, Lemma 4.9]. In the infinite dimensional case here, we also have similar story. The ***energy of Floer cylinder*** $u \in C^\infty(\mathbb{R}\times\mathbb{S}^1, M)$ is defined by

$$\begin{aligned} E(u) :=& \int_{-\infty}^{+\infty} \|\partial_s u\|_J^2 \, ds \\ =& \frac{1}{2}\int_{-\infty}^{+\infty} \left(\|\partial_s u\|_J^2 + \|\partial_t u - X_{H_t}(u)\|_J^2 \right) ds. \end{aligned}$$

Using equations (6) and (7), we can further express the energy as

$$E(u) = \int_{-\infty}^{+\infty} \|\operatorname{grad} \mathscr{A}_H\|_J^2 \, ds = -\int_{-\infty}^{+\infty} \frac{d}{ds} \mathscr{A}_H(u(s)) ds.$$

Theorem 3.1 (Asymptotic Convergence). *Let $u \in C^\infty(\mathbb{R}\times\mathbb{S}^1, M)$ be a solution of Floer equation (7). Then $E(u) < +\infty$ iff. there exists $\gamma^\pm \in \mathscr{P}_0(H)$ such that in C^∞-topology*

$$\lim_{s\to\pm\infty} u(s,\cdot) = \gamma^\pm, \qquad \lim_{s\to\pm\infty} \partial_s u(s,\cdot) = 0.$$

In particular,

$$E(u) = \mathscr{A}_H(\gamma^-) - \mathscr{A}_H(\gamma^+),$$

and $E(u) = 0$ when u is a constant flow line, i.e. $u(s,\cdot) = \gamma^- = \gamma^+ \in \mathscr{P}_0(H)$ $\forall\, s \in \mathbb{R}$.

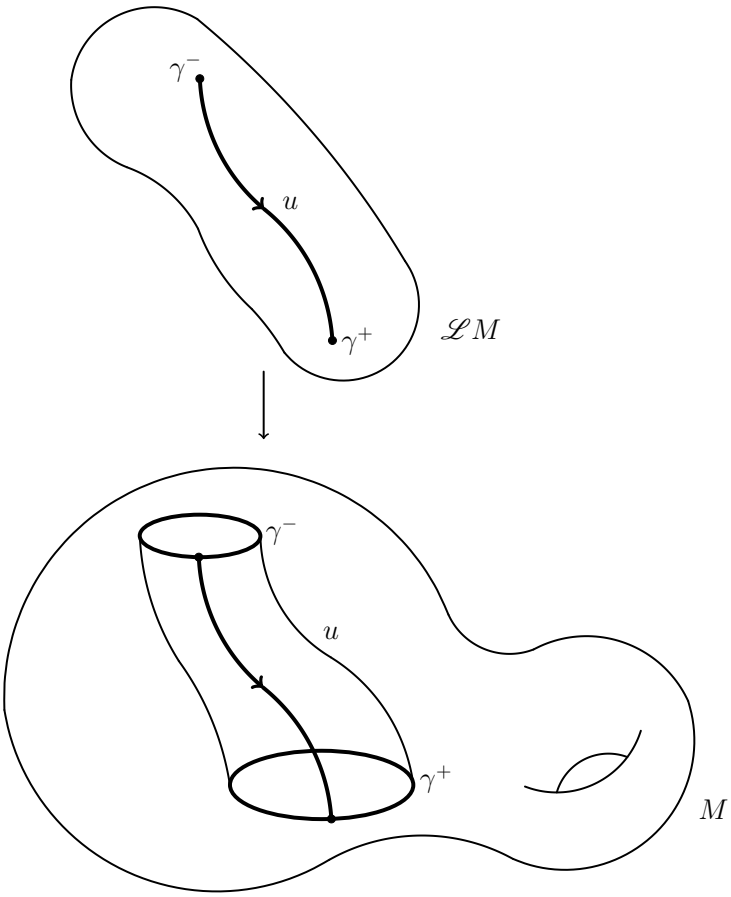


Figure 2: Two viewpoints of a negative gradient flow line u of the symplectic action functional $\mathscr{A}_H$ connecting two critical points $\gamma^\pm \in \operatorname{Crit} \mathscr{A}_H$.

See Figure 2 for the illustration of such asymptotic convergence. We refer to [AD14, §6.5b] for the detail discussion and proof.

This implies the space of Floer cylinders of finite energy

$$\widetilde{\mathcal{M}} := \{u \in C^\infty(\mathbb{R} \times \mathbb{S}^1, M) : u \text{ solves (7) with } E(u) < +\infty\}$$

is decomposed into subspaces

$$\widetilde{\mathcal{M}} = \bigcup_{\gamma^-, \gamma^+ \in \mathscr{P}_0(H)} \widetilde{\mathcal{M}}(\gamma^-, \gamma^+)$$

where

$$\widetilde{\mathcal{M}}(\gamma^-, \gamma^+) := \{u \in C^\infty(\mathbb{R} \times \mathbb{S}^1, M) : u \text{ solves (7) with } \lim_{s \to \pm\infty} u(s, \cdot) = \gamma^\pm\}.$$

The space $\widetilde{\mathcal{M}}$ resp. $\widetilde{\mathcal{M}}(\gamma^-, \gamma^+)$ is called the ***moduli space of parameterized negative gradient flow lines of*** $\mathscr{A}_H$ of finite energy resp. with asymptotics $\gamma^\pm$, as in analogy to the moduli space of negative gradient flow lines of Morse function in the finite dimensional case.

We now analyse local converging behavior for a sequence of Floer cylinders in analogy to the local convergence of sequence of negative gradient flow lines in finite dimensional Morse theory, c.f. [FN20, §4.4].
Then we have the following local convergence result.

Theorem 3.2 (Local Convergence). *Let $H \in C^\infty(M \times \mathbb{S}^1)$ be a nondegenerate Hamiltonian on a symplectically aspherical manifold (M, ω). For a sequence of Floer cylinders of finite energy $(u_k)_{k\in\mathbb{N}} \subset \widetilde{\mathcal{M}}$ is, there exists a subsequence $(u_{k_j})_{j\in\mathbb{N}}$ and a finite energy Floer cylinder $u \in \widetilde{\mathcal{M}}$ such that*

$$u_{k_j} \xrightarrow{C^\infty_{\mathrm{loc}}} u \qquad \text{as } j \to \infty,$$

i.e. for every compact subset $K \subset \mathbb{R} \times \mathbb{S}^1$, $u_{k_j}|_K \xrightarrow{j\to\infty} u|_K$ in $C^\infty(K, M)$ uniformly.

Remark 3.1. The theorem implies that the moduli space $\widetilde{\mathcal{M}}$ is compact in C^∞_{loc}-topology.

The geometric assumption of symplectic asphericity is crucial as it will enforce L^∞-bound on the derivatives of Floer cylinders of finite energy, i.e. $\exists\, C > 0$ such that for any $u \in \widetilde{\mathcal{M}}$ the following holds:

$$\|\partial_s u\|_{L^\infty(\mathbb{R}\times\mathbb{S}^1, M)} \leq C.$$

This will give the uniform equicontinuity of elements in $\widetilde{\mathcal{M}} \subset C^0_{\mathrm{loc}}(\mathbb{R} \times \mathbb{S}^1, M)$. Thus, by the Arzelà–Ascoli theorem, there is a converging subsequence $(u_{k_j}) \subseteq (u_k)$ in $C^0_{\mathrm{loc}}(\mathbb{R} \times \mathbb{S}^1, M)$. Denoting by u_∞ the limit object in $C^0(\mathbb{R} \times \mathbb{S}^1, M)$, and a ball-shaped open cover (B_j) of $\mathbb{C}^\times \cong \mathbb{R} \times \mathbb{S}^1$ with a partition of unity (β_j) subordinating to it. Multiplying these cut-off functions β_j with u_k, we obtained a family of compact supported smooth maps $\bar{u}^j_k \in C^\infty_c(\mathbb{C}^\times, M)$. Differentiating the Floer equation (7) yields expression of the Laplacian $\Delta\bar{u}^j_k$ (i.e. second derivatives) in terms of polynomial of first derivatives $\partial_s u_k$ and $\partial_t u_k$ with C^∞-coefficient. The L^∞-bound on $\partial_s u_k$ then gives rise L^∞-bound on $\Delta\bar{u}^j_k$, and hence due to compact support one gets L^p-bound on $\Delta\bar{u}^j_k$ for all $p \in [1, \infty]$. The Calderon-Zygmund inequality implies that $\bar{u}^j_k$ is of $W^{2,p}$-class for all $p \in (1, \infty)$. Further by Morrey inequality, the first derivatives $\partial_s\bar{u}^j_k$ and $\partial_t\bar{u}^j_k$ are of $C^{0,\alpha}$-class for $\alpha \in (0, 1)$. Again by Arzelà–Ascoli theorem, $\partial_s\bar{u}^j_k$ and $\partial_t\bar{u}^j_k$ have subsequences converging to $\bar{u}^j_\infty \in C^0_c(\mathbb{C}^\times, M)$, thus $\bar{u}^j_\infty \in C^1_c(\mathbb{C}^\times, M)$. This means $u_k|_{B_j} \xrightarrow{C^1} \bar{u}^j_\infty$, and therefore $u_\infty \in C^1(\mathbb{C}^\times, M)$. Iterating above procedure, one eventually concludes that $u_k \xrightarrow{C^\infty_{\mathrm{loc}}} u_\infty$. Such procedure is known as *elliptic bootstrap*. Consequently, the C^k_{loc}-topologies on $\widetilde{\mathcal{M}}$ for all $k \in \mathbb{N}$

coincides. These analysis will be carried out in detail in Section 3.4 for the *bubbling*.

Next, we discuss the convergence behavior w.r.t. the stratification of $\widetilde{\mathcal{M}}$, namely the strata $\widetilde{\mathcal{M}}(\gamma^-, \gamma^+)$. Assume we have a sequence $(u_k)_{k\in\mathbb{N}}$ of negative gradient flow lines of the symplectic action functional $\mathscr{A}_H$ or Floer cylinders with asymptotic convergence

$$\lim_{s\to\pm\infty} u_k(s,\cdot) = \gamma^\pm \in \mathscr{P}_0(H) \qquad \forall k \in \mathbb{N}.$$

See Figure 3. For such a sequence $(u_k)_{k\in\mathbb{N}}$ of Floer cylinders, we have the uniform energy bound:

$$E(u_k) = \mathscr{A}_H(\gamma^-) - \mathscr{A}_H(\gamma^+) < +\infty \qquad \forall\, k \in \mathbb{N}.$$

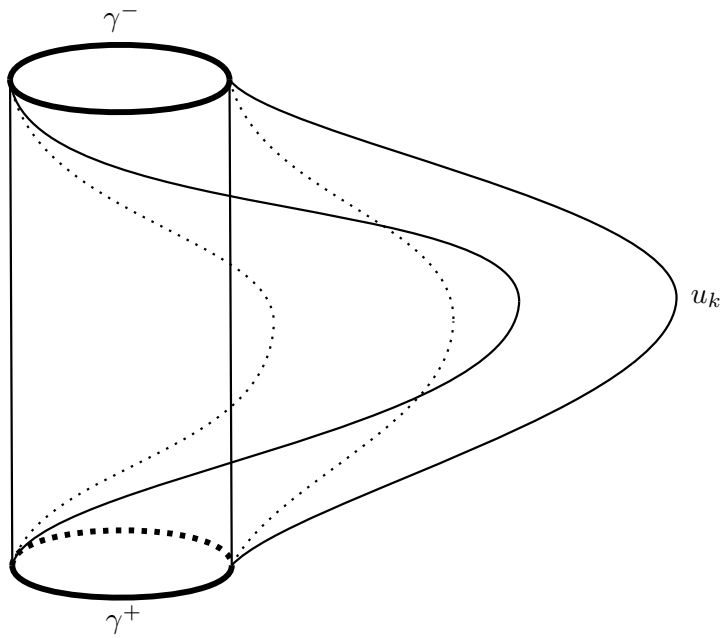


Figure 3: A sequence (u_k) of negative gradient flow lines asymptotically converging to the same critical points $\gamma^\pm$ of the symplectic action functional $\mathscr{A}_H$.

In the finite dimensional Morse theory, the moduli space of gradient flow lines with two fixed asymptotics will be noncompact in general, and it can be compactified by considering broken flow lines as limit object under the notion of Floer-Gromov convergence, c.f. [FN20, §4.7]. For Floer cylinders, the same happens, and we state the Floer-Gromov convergence theorem below.

Theorem 3.3 (Floer-Gromov Convergence). *Let $(u_k)_{k\in\mathbb{N}}$ be a sequence of Floer cylinders in $\widetilde{\mathcal{M}}(\gamma^-, \gamma^+)$ with $\gamma^- \neq \gamma^+$. Then there is a subsequence*

$(u_{k_j})_{j\in\mathbb{N}} \subset \widetilde{\mathcal{M}}(\gamma^-,\gamma^+)$, *a sequences of real numbers* $(r_j^i)_{i\in\{1,\dots,m\},j\in\mathbb{N}} \subset \mathbb{R}$, *and a* ***broken Floer cylinder*** *which is a tupule* $(u^i)_{i\in\{1,\dots,m\}}$ *where* u^i *is nonconstant Floer cylinder for every* i *with asymptotics*

$$\lim_{s\to+\infty} u^i(s,\cdot) = \lim_{s\to-\infty} u^{i+1}(s,\cdot) = \gamma^i \in \mathscr{P}_0(H),$$

in particular $\gamma^0 = \gamma^-, \gamma^m = \gamma^+$, *such that*

$$r_j^i \cdot u_{k_j} \xrightarrow{C^\infty_{\mathrm{loc}}} u^i \qquad \text{as } j \to \infty$$

where $r_j^i \cdot u_{k_j}(s,t) = u_{k_j}(s + r_j^i, t)$ *is the time-shift. We say that* (u_k) *is Floer-Gromov converging to* $(m-1)$*-broken Floer cylinder* $(u^i)_{i\in\{1,\dots,m\}}$ *connecting* γ^- *to* γ^+.

Remark 3.2. The limit broken Floer cylinder is actually unique up to time-shift. The Floer-Gromov convergence defines a topology on the space $\overline{\widetilde{\mathcal{M}}(\gamma^-,\gamma^+)}$ which is $\widetilde{\mathcal{M}}(\gamma^-,\gamma^+)$ including limiting broken Floer cylinders, called the Floer-Gromov topology. See [FN20, §4.11]. In such way, the moduli space is compactified into a space $\overline{\widetilde{\mathcal{M}}(\gamma^-,\gamma^+)}$ that is sequentially compact. In lateral exploration, we will realize the compactified moduli space is a smooth finite dimensional Banach manifold, thus it is compact w.r.t. the metric topology. This compactness result is called the ***Floer-Gromov compactness***.

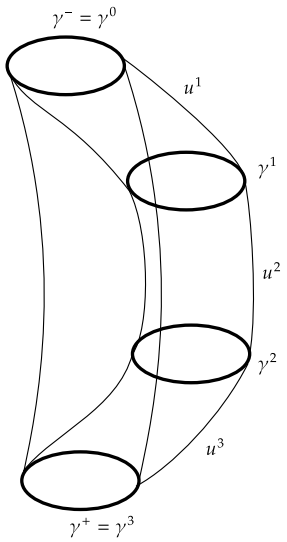


Figure 4: Floer-Gromov convergence towards a broken Floer cylinder with two breaks.

3.2 Bubbling Phenomenon

The negative gradient flow equation (7) also makes sense on general symplectic manifold that is not necessary symplectically aspherical, even though the symplectic action functional (1) is not well-defined (it can be however defined on certain covering sapce of $\mathscr{L}_0 M$ which we will encounter later). A possible limit scenario is depicted in Figure 5. The negative gradient flow lines might break in the limit giving rise to broken gradient flow lines. But in contrast to the finite dimensional setting, a *bubbling phenomenon* occurs, that is, the broken Floer cylinder in the limit might have nonconstant pseudoholomorphic spheres attached. The reason behind such difference is that in the finite-dimensional case, compactness follows from the uniform bound of derivatives of the gradient flow lines due to the L^∞-bound on the gradient of Morse function by applying the theorem of Arzelà–Ascoli. Whereas in the infinite-dimensional case, we do not have in general such a uniform bound on both of the partial derivatives due to lack of L^∞-bound on the L^2-gradient of the symplectic action functional. This is mainly because the space $\mathscr{L}_0 M$ is not compact. The analysis of this limit behaviour is very hard from an analytical point, that is, by just studying the solutions of the Floer equation analitically. However, the geometric assumption of symplectical asphericity excludes the existence of such limit pseudoholomorphic curves. Indeed, we have the following result.

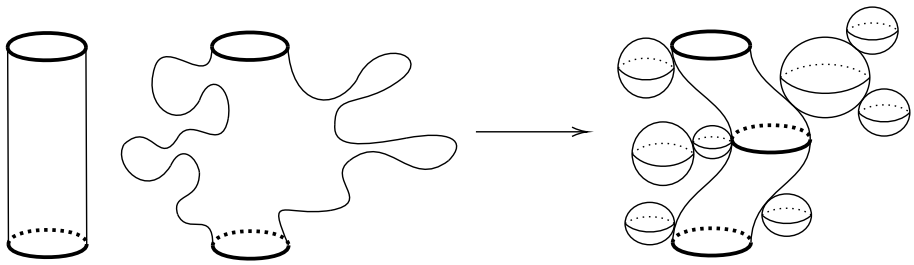


Figure 5: The bubbling phenomenon of a sequence of solutions of the Floer equation.

Lemma 3.1. *Let $v\colon (\Sigma, j) \to (M, J)$ be a non-constant pseudoholomorphic curve for a symplectic manifold (M, ω) with ω-compatible almost complex structure J. Then the symplectic area of v is strictly positive:*

$$\int_\Sigma v^*\omega > 0.$$

Proof. Recall J-holomorphic curve satisfies the non-linear Cauchy-Riemann equation

$$\bar{\partial}_J v := \frac{1}{2}(dv + J \circ v \circ j) = 0.$$

Around a point $z \in \Sigma$, we choose a local holomorphic coordinate $\varphi : \Sigma \supset U \to V \subset \mathbb{C}$ s.t. $\varphi(z) = z = s + it \in \mathbb{C}$ w.r.t. standard complex structure i on $\mathbb{C}$, and put $\tilde{v} := v \circ \varphi^{-1} : V \to M$. Then we compute the local symplectic area:

$$\int_U v^*\omega = \int_V \tilde{v}^*\omega = \int_V \omega_{\tilde{v}(z)}(\partial_s\tilde{v}(z), \partial_t\tilde{v}(z))ds \wedge dt$$
$$\overset{\partial_s\tilde{v}+J\partial_t\tilde{v}=0}{=} \int_V \omega_{\tilde{v}(z)}(\partial_s\tilde{v}(z), J(\tilde{v}(z))\partial_s\tilde{v})ds \wedge dt.$$

We know that $g_J(\partial_s\tilde{v}, \partial_s\tilde{v}) := \omega(\partial_s\tilde{v}, J\partial_s\tilde{v}) \geq 0$ since J is ω-compatible (i.e. g_J is a Riemannian metric). By assumption that v is non-constant, hence its derivatives can not be globally vanishing, there must exist some point $z \in \Sigma$ so that the surrounding local symplectic area is strictly greater than zero. We therefore conclude that non-constant J-holomorphic curve has strictly positive symplectic area. □

Remark 3.3. We see that the symplectic area of pseudoholomorphic curve is exactly the energy of it. Actually, there is a finer result for pseudoholomorphic spheres: For every ω-compatible almost complex structure J, there exists a constant $\hbar = \hbar(J) > 0$ such that every J-holomorphic sphere $v : \mathbb{S}^2 \to M$ has uniform energy lower bound $E(v) \geq \hbar$.

In summary, the strategy to tackle the compactness of the moduli space of negative gradient flow lines of the symplectic action functional is the following. As the loop space is not compact, gradient flow lines can escape to infinity. Since the underlying manifold is compact, the image of Floer cylinders must be bounded, thus the derivatives should explode. We will show by contradiction, that if the derivatives explode, there must exist a nonconstant pseudoholomorphic sphere. Whereas Lemma 3.1 leads to contradiction to the symplectic asphericity of (M, ω).

In the following, we dedicate to show that pseudoholomorphic bubble occurs in the limit of Floer cylinders on general symplectic manifold when there is no uniform bound for the partial derivatives.

Theorem 3.4 (Bubbling). *Let (M, ω) be a compact symplectic manifold and (u_k) a sequence of solutions of the Floer equation (7) for some $H \in C^\infty(M \times \mathbb{S}^1)$. Suppose that there exists a constant $C > 0$ such that for every smooth family $(J_t)_{t\in\mathbb{S}^1}$ of ω-compatible almost complex structures, the energy is finite, i.e.*

$$E(u_k) := \int_{-\infty}^{+\infty} \|\partial_s u_k\|_J^2 \, ds := \int_{-\infty}^{+\infty} \int_0^1 g_{J_t}(\partial_s u_k, \partial_s u_k) \, dtds \leq C \qquad \forall\, k \in \mathbb{N},$$

and there exists a sequence (s_k, t_k) in $\mathbb{R} \times \mathbb{S}^1$ such that

$$\|\partial_s u_k(s_k, t_k)\|_{g_{J_{t_k}}} \xrightarrow{k\to\infty} \infty.$$

Then there exists $v : \mathbb{S}^2 \to M$ *nonconstant* J*-holomorphic* ($J := J_{t_\infty}$) *such that*

$$u_k(s_k, t_k) \xrightarrow{k\to\infty} v(0)$$

where $0 \in \mathbb{S}^2 \cong \mathbb{C} \cup \{\infty\}$.

Remark 3.4. One can show that, due to the uniform upper bound on the energy of (u_k) and the uniform lower bound on the energy of pseudoholomorphic spheres, there exist at most finitely many limit points on the cylinder, e.g. $\{z_\infty^1, \dots, z_\infty^\ell\} \subset \mathbb{R} \times \mathbb{S}^1$, on which the derivatives of (u_k) blow up and hence bubbling off pseudoholomorphic spheres $\{v^1, \dots, v^\ell\}$ in above fashion. By Theorem 3.2, there is a Floer cylinder u such that $u_k \xrightarrow{C^\infty_{\text{loc}}} u$ as $k \to \infty$ on $\mathbb{R} \times \mathbb{S}^1 \backslash \{z_\infty^1, \dots, z_\infty^\ell\}$. The points $v^1(\infty), \dots, v^\ell(\infty) \in M$ are the attaching points of the pseudoholomorphic spheres to the limit object of (u_k), which are called the nodal points. Eventually, the limit object under Floer-Gromov convergence is a broken Floer cylinder attached with bubble trees which is called the stable Floer cylinder. See the right hand side of the Figure 5. Suppose there is a $(m-1)$-broken Floer cylinder $(u^i)_{i=1,\dots,m}$ in the limit, the energy of broken Floer cylinder is bounded from above by

$$\sum_{i=1}^{m} E(u^i) \leq \limsup_{k\to\infty} E(u_k) - \ell \inf_{t\in\mathbb{S}^1} \hbar(J_t).$$

The idea of the proof is to rescale the sequence (u_k). Set

$$m_k := \|\partial_s u_k(s_k, t_k)\|_{g_{J_{t_k}}} \qquad \text{and} \qquad v_k(\sigma, \tau) := u_k\left(\frac{\sigma}{m_k} + s_k, \frac{\tau}{m_k} + t_k\right)$$

for all $(\sigma, \tau) \in \mathbb{R} \times \mathbb{S}^1$. Then

$$\begin{aligned} m_k \partial_\sigma v_k(\sigma, \tau) &= \partial_s u_k\left(\frac{\sigma}{m_k} + s_k, \frac{\tau}{m_k} + t_k\right) \\ m_k \partial_\tau v_k(\sigma, \tau) &= \partial_t u_k\left(\frac{\sigma}{m_k} + s_k, \frac{\tau}{m_k} + t_k\right), \end{aligned}$$

and plugging this into the Floer equation (7) yields that v_k satisfies

$$\partial_\sigma v_k(\sigma, \tau) + J_{\frac{\tau}{m_k}+t_k} \partial_\tau v_k(\sigma, \tau) = \frac{1}{m_k} J_{\frac{\tau}{m_k}+t_k} X_{H_{\frac{\tau}{m_k}+t_k}}(v_k(\sigma, \tau))$$

for all $k \in \mathbb{N}_0$. If there exists a solution v of the Floer equation such that $v_k \to v$, then v satisfies

$$\partial_\sigma v(\sigma, \tau) + J_{t_\infty} \partial_\tau v(\sigma, \tau) = 0$$

as $k \to \infty$ for some $t_\infty \in \mathbb{S}^1$. Consequently, $v \colon \mathbb{C} \to M$ is a J_{t_∞}-holomorphic plane. Using the identification $\mathbb{S}^2 \cong \mathbb{CP}^1 \cong \mathbb{C} \cup \{\infty\}$, we extend v to a pseudoholomorphic sphere via a procedure called *removal of singularities*. The rescaling argument can be thinked of as looking at smaller and smaller balls on the

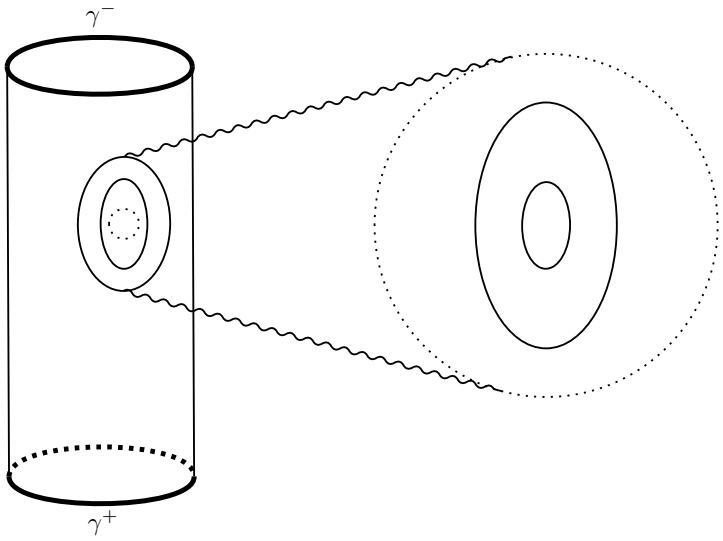


Figure 6: Looking at the sequence (u_k) of the Floer equation with a magnifying glass via rescaling.

Floer cylinder via a microscope. See Figure 6. The subtlety in the above argument is choosing the right rescaling. Indeed, a necessary condition for convergence modulo subsequences of (v_k) is that the derivatives are uniformly bounded. This might fail for a too small rescaling. On the other hand, if the rescaling is too big, the derivatives might become vanishing and the resulting pseudoholomorphic sphere would be constant. As the result stated in the Theorem 3.4, the rescaling procedure that we are going to carry out only sees one bubble, not the entire bubble tree structure, since we only care about the existence of bubble. Namely we only obtain a bubble without knowing the position it lands: It could be a bubble attaching to a Floer cylinder or to another bubble in the limit object. Therefore, such procedure is called the *hard rescaling*. To realize the bubble tree structure, one needs to implement the *soft rescaling*, which is beyond the scope of this note.

We divide the proof of Theorem 3.4 into several technical steps in the following sections.

3.3 Rescaling

To manifest such a delicate rescaling, we need the following technical lemma from Helmut Hofer.

Lemma 3.2 (Hofer). *Assume (X, d) is a complete metric space, $f : X \to \mathbb{R}_{\geq 0}$ is continuous. Then for every $x \in X$ and $\delta > 0$, $\exists\, z \in X$ and $0 < \varepsilon \leq \delta$ s.t.*

$$d(x, z) < 2\delta, \qquad \sup_{y \in B_\varepsilon(z)} f(y) \leq 2f(z), \qquad \varepsilon f(z) \geq \delta f(x),$$

where $B_\varepsilon(z) := \{y \in X : d(y,z) \le \varepsilon\}$ *as the closed* ε*-ball around* z*.*

Proof. We prove the lemma by contradiction: Suppose the statement is wrong.

Claim. $\exists\, \{x_k\}_{k\in\mathbb{N}} \subset X$ s.t. $x_0 = x$, $d(x_k, x_{k+1}) \le \frac{\delta}{2^k}$, $f(x_{k+1}) > 2f(x_k)$.

We employ the induction procedure to pove the claim: The assumption that the statement is wrong implies that it is particularly wrong for the case where $z = x = x_0$, $\varepsilon = \delta$. This implies that

$$\sup_{y\in B_\delta(x_0)} f(y) > 2f(x_0),$$

and further implies that $\exists\, x_1 \in X$ s.t. $d(x_0, x_1) \le \delta$, $f(x_1) > 2f(x_0)$.

Induction hypothesis: Such $x_0, \dots, x_k$ $(k \ge 1)$ have been constructed that the statement is wrong for them.

This implies that $d(x, x_k) < 2\delta$, $f(x_k) > 2^k f(x_0)$. The statement is wrong for $z = \xi_k$, $\varepsilon = \frac{\delta}{2^k}$.

$$\Rightarrow \sup_{y\in B_{\frac{\delta}{2^k}}(x_k)} f(y) > 2f(x_k).$$

$$\Rightarrow \exists\, x_{k+1} \in X \text{ s.t. } d(x_k, x_{k+1}) \le \frac{\delta}{2^k},\ f(x_{k+1}) > 2f(x_k).$$

This terminates the induction step, and thus proves the claim.

$B_{2\delta}(x)$ is closed in complete metric space, so $\{x_k\}_{k\in\mathbb{N}}$ is a Cauchy sequence $\Rightarrow$ $\exists\, y \in B_{2\varepsilon}(x)$ s.t. $x_k \xrightarrow{k\to\infty} y$. f continuous $\Rightarrow \lim_{k\to\infty} f(x_k) = f(y) < +\infty$. Since $f(x_1) > 2f(x_0) \ge 0$, then $f(x_k) > 2^{k-1} f(x_1) \Rightarrow f(x_k) \xrightarrow{k\to\infty} +\infty$. This leads to contradiction! □

Now we apply Hofer's lemma on Floer cylinder to obtain the non-constant pseudoholomorphic plane on it.

Lemma 3.3. *There exists a sequence* (R_k) *in* $(0, +\infty)$ *with* $R_k \to +\infty$*, a sequence* (J_k) *of* ω*-compatible almost complex structures on* (M, ω) *and a sequence* $v_k \in C^\infty(B_{R_k}(0), M)$ *such that*

$$\lim_{k\to\infty} \|\partial_\sigma v_k + J_k \partial_\tau v_k\|_{L^\infty} = 0 \qquad \textit{and} \qquad 1 \le \|\partial_\sigma v_k\|_{L^\infty} \le 2.$$

Proof. Abbreviate $z_k := (s_k, t_k) \in \mathbb{R} \times \mathbb{S}^1$. Apply Hofer's Lemma 3.2 with

$$x = z_k, \qquad \delta_k := \frac{1}{\sqrt{m_k}} \qquad \text{and} \qquad f := \|\partial_s u_k\|.$$

Note that since M is finite dimensional compact manifold, the Riemannian metrics $(g_{J_t})_{t\in\mathbb{S}^1}$ are all (locally) equivalent, so we denote all the induced equivalent norms here simply as $\|\cdot\|$ without explicit subscript. Thus there exists $z_k' = (s_k', t_k') \in \mathbb{R}\times\mathbb{S}^1$ and $0 < \varepsilon_k \leq \delta_k$ such that

$$\sup_{z\in B_{\varepsilon_k}(z_k')} \|\partial_s u_k(z)\| \leq 2\,\|\partial_s u_k(z_k')\|$$

and

$$\varepsilon_k\,\|\partial_s u_k(z_k')\| \geq \delta_k\,\|\partial_s u_k(z_k)\| = \sqrt{m_k}.$$

Rescale with $m_k' := \|\partial_s u_k(z_k')\|$, set $R_k := \varepsilon_k m_k'$ and

$$v_k(\sigma,\tau) := u_k\left(\frac{\sigma}{m_k'} + s_k', \frac{\tau}{m_k'} + t_k'\right) \qquad \forall (\sigma,\tau)\in B_{R_k}(0).$$

As (u_k) is a solution to the Floer equation, v_k satisfies the equation

$$\partial_\sigma v_k(\sigma,\tau) + J_{\frac{\tau}{m_k'}+t_k'}\partial_\tau v_k(\sigma,\tau) = \frac{1}{m_k'} J_{\frac{\tau}{m_k'}+t_k'} X_{H_{\frac{\tau}{m_k'}+t_k'}}(v_k(\sigma,\tau)).$$

As $m_k' \geq m_k \to \infty$, we conclude

$$\lim_{k\to\infty} \|\partial_\sigma v_k + J_k\partial_\tau v_k\|_{L^\infty} = 0$$

for $J_k := J_{\frac{\tau}{m_k'}+t_k'}$. Moreover, we estimate

$$\begin{aligned}
\|\partial_\sigma v_k\|_{L^\infty} &= \sup_{z\in B_{R_k}(0)} \|\partial_\sigma v_k(z)\| \\
&= \frac{1}{m_k'} \sup_{z\in B_{\varepsilon_k}(z_k')} \|\partial_s u_k(z)\| \\
&\leq \frac{2}{m_k'}\,\|\partial_s u_k(z_k')\| \\
&= 2,
\end{aligned}$$

and

$$m_k'\partial_\sigma v_k(0,0) = \partial_s u_k(z_k').$$

This concludes the proof of the lemma. □

3.4 Elliptic Bootstrapping

The aim of this section is to show that there exists a subsequence of (v_k) from Lemma 3.3, an ω-compatible almost complex structure J_{t_∞} on (M,ω) for some $t_\infty \in \mathbb{S}^1$ and a J_{t_∞}-holomorphic plane $v\colon \mathbb{C}\to M$ with

$$v_k \xrightarrow{C^\infty_{\text{loc}}} v.$$

Lemma 3.4. *Let (v_k) be the sequence constructed in Lemma 3.3. Then there exists $v_\infty \in C^0(\mathbb{C}, M)$ such that*

$$v_k \xrightarrow{C^0_{\mathrm{loc}}} v_\infty$$

up to subsequences.

Proof. By Lemma 3.3 we get

$$\|\partial_\sigma v_k\|_{L^\infty} \leq 2 \tag{8}$$

Moreover, v_k solves

$$\begin{aligned}\partial_\sigma v_k(\sigma,\tau) + J_{\frac{\tau}{m_k'}+t_k'}\partial_\tau v_k(\sigma,\tau) &= \frac{1}{m_k'} J_{\frac{\tau}{m_k'}+t_k'} X_{H_{\frac{\tau}{m_k'}+t_k'}}(v_k(\sigma,\tau)) \\ &= -\frac{1}{m_k'}\operatorname{grad}_{J_{\frac{\tau}{m_k'}+t_k'}} H_{\frac{\tau}{m_k'}+t_k'}(v_k(\sigma,\tau)) \end{aligned} \tag{9}$$

for all $(\sigma,\tau) \in B_{R_k}(0)$, where the gradient $\operatorname{grad}_{J_t} H_t$ is with respect to the Riemannian metric $\omega(\cdot, J_t\cdot)$ for all $t \in \mathbb{R}$. Thus by compactness of M we conclude from (9) and (8) that there exists a constant $C > 0$ with

$$\|\partial_\tau v_k\|_{L^\infty} \leq C.$$

For every fixed $R > 0$, we choose $N_R \in \mathbb{N}$ such that $R_k \geq R$ for all $k \geq N_R$ and consider the restrictions $v_k|_{B_R(0)}$ for all $k \geq N_R$. As the derivatives of v_k are uniformly bounded and M is compact, we conclude that $v_k|_{B_R(0)}$ is of Sobolev class

$$W^{1,\infty}(B_R(0), M) := \{f \in W^{1,\infty}(B_R(0), \mathbb{R}^{4n+1}) : f(B_R(0)) \subseteq M\}$$

for all $k \geq N_k$, where we consider $M^{2n} \hookrightarrow \mathbb{R}^{4n+1}$ via the Whitney embedding. Thus by Morrey's inequality [Bre11, Corollary 9.14], every $v_k|_{B_R(0)}$ is of Hölder class $C^{0,1}(B_R(0), M)$ (or 1-Lipschitz continuous) and hence $v_k|_{B_R(0)}$ is equicontinuous. As M is compact, the Arzelá-Ascolis theorem implies that there exists $v^R \in C^0(B_R(0), M)$ such that

$$v_k|_{B_R(0)} \xrightarrow{C^0} v^R$$

up to a subsequence. Choose a subsequence (k_j^1) with $k_j^1 \geq N_1$ for all $j \in \mathbb{N}$ and such that there exists $v^1 \in C^0(B_1(0), M)$ with

$$v_{k_j^1}|_{B_1(0)} \xrightarrow{C^0} v^1$$

as $j \to \infty$. Choose a subsequence (k_j^2) of (k_j^1) with $k_j^2 \geq N_2$ for all $j \in \mathbb{N}$ and such that there exists $v^2 \in C^0(B_2(0), M)$ with

$$v_{k_j^2}|_{B_2(0)} \xrightarrow{C^0} v^2$$

as $j \to \infty$. This implies $v^2|_{B_1(0)} = v^1$. Inductively, choose a subsequence $(k_j^{\ell+1})$ of (k_j^ℓ) for all $\ell \in \mathbb{N}$ with $k_j^{\ell+1} \geq N_{\ell+1}$ for all $j \in \mathbb{N}$ and such that there exists $v^{\ell+1} \in C^0(B_{\ell+1}(0), M)$ with

$$v_{k_j^{\ell+1}}|_{B_{\ell+1}(0)} \xrightarrow{C^0} v^{\ell+1}$$

as $j \to \infty$. Finally, we employ the diagonal sequence trick [RS81, Theorem 1.24]: We choose the diagonal subsequence $(k_j^\ell)_{\ell=j} \subset (k_j)$, and denote it as $(d_j) := (k_j^j)$. Then the subsequence $(v_{d_j}) \subset (v_{k_j}) \subset (v_k)$ is converging uniformly on $B_R(0)$ as $j \to \infty$. Define $v_\infty \in C^0(\mathbb{C}, M)$ to be the limit $\lim_{j\to\infty} v_{d_j}|_{B_R(0)}$, and note $v_\infty|_{B_\ell(0)} = v^\ell$. Therefore, we conclude that up to subseqeunce,

$$v_k \xrightarrow{C^0_{\text{loc}}} v_\infty$$

as $k \to \infty$. □

Lemma 3.5. *Denote by $v_\infty \in C^0(\mathbb{C}, M)$ the map constructed in Lemma 3.4. Then actually $v_\infty \in C^\infty(\mathbb{C}, M)$ is nonconstant, with*

$$v_k \xrightarrow{C^\infty_{\text{loc}}} v_\infty \qquad \text{and} \qquad \partial_\sigma v_\infty + J_{t_\infty}\partial_\tau v_\infty = 0$$

for some $t_\infty \in \mathbb{S}^1$. Thus v_∞ is a nonconstant pseudoholomorphic plane.

Proof. Fix sequences $(z_j) \subseteq \mathbb{C}$ and $r_j \subseteq (0, +\infty)$ such that $v_\infty|_{B_{4r_j}(z_j)}$ is contained in a chart U_j of M and $\bigcup_{j\in\mathbb{N}} \mathring{B}_{r_j}(z_j) = \mathbb{C}$. Moreover, for all $j \in \mathbb{N}$ there exists $N_j \in \mathbb{N}$ such that $v_k|_{B_{2r_j}(z_j)} \subseteq U_j$ for all $k \geq N_j$. Fix smooth bump functions $\beta_j \in C^\infty(\mathbb{C}, [0,1])$ for $\bar{B}_{r_j}(z_j)$ and supported in $B_{2r_j}(z_j)$. Define

$$\bar{v}_k^j := \beta_j v_k \in C_c^\infty(\mathbb{C}, \mathbb{R}^{2n})$$

for all $k, j \in \mathbb{N}$. We compute

$$\Delta\bar{v}_k^j = (\Delta\beta_j)v_k + 2\partial_\sigma\beta_j\partial_\sigma v_k + 2\partial_\tau\beta_j\partial_\tau v_k + \beta_j\Delta v_k. \tag{10}$$

To compute Δv_k, we differentiate (9) in charts. Applying ∂_σ we get

$$\partial_\sigma^2 v_k(\sigma,\tau) + DJ_{\frac{\tau}{m_k'}+t_k'}\partial_\sigma v_k(\sigma,\tau)\partial_\tau v_k(\sigma,\tau) + J_{\frac{\tau}{m_k'}+t_k'}\partial_\sigma\partial_\tau v_k(\sigma,\tau)$$

is equal to

$$-\frac{1}{m_k'} D\operatorname{grad}_{J_{\frac{\tau}{m_k'}+t_k'}} H_{\frac{\tau}{m_k'}+t_k'}(v_k(\sigma,\tau))\partial_\sigma v_k(\sigma,\tau).$$

Applying ∂_τ we get

$$\partial_\tau\partial_\sigma v_k(\sigma,\tau) + J_{\frac{\tau}{m_k'}+t_k'}\partial_\tau^2 v_k(\sigma,\tau) + DJ_{\frac{\tau}{m_k'}+t_k'}(\partial_\tau v_k(\sigma,\tau))^2 + \frac{1}{m_k'}\dot{J}_{\frac{\tau}{m_k'}+t_k'}\partial_\tau v_k(\sigma,\tau)$$

is equal to

$$-\frac{1}{m_k'} D \operatorname{grad}_{J_{\frac{\tau}{m_k'}+t_k'}} H_{\frac{\tau}{m_k'}+t_k'}(v_k(\sigma,\tau))\partial_\tau v_k(\sigma,\tau)$$
$$-\frac{1}{m_k'} \partial_\tau \operatorname{grad}_{J_{\frac{\tau}{m_k'}+t_k'}} H_{\frac{\tau}{m_k'}+t_k'}(v_k(\sigma,\tau)).$$

Hence

$$\Delta v_k = P(\partial_\sigma v_k, \partial_\tau v_k), \tag{11}$$

where P is a polynomial of degree 2 with C^∞-coefficients depending on u_k and τ. Consequently, we have that

$$\Delta \bar{v}_k^j = P(\partial_\sigma \bar{v}_k^j, \partial \bar{v}_k^j)$$

is also a polynomial of degree 2 by (10). As $\partial_\sigma v_k$ and $\partial_\tau v_k$ are uniformly bounded, we conclude that $\Delta \bar{v}_k^j$ is uniformly bounded. Thus $\|\Delta \bar{v}_k^j\|_{L^p}$ is uniformly bounded for all $p \in [1, +\infty]$ as $\Delta \bar{v}_k^j$ is compactly supported. Thus by the Calderon–Zygmund inequality [MS12, Corollary B.2.7], we conclude that $\|\bar{v}_k^j\|_{W^{2,p}}$ is uniformly bounded for all $1 < p < +\infty$. In particular,

$$\|\partial_\sigma \bar{v}_k^j\|_{W^{1,p}} \qquad \text{and} \qquad \|\partial_\tau \bar{v}_k^j\|_{W^{1,p}}$$

are uniformly bounded. Thus by Morrey's inequality, $\partial_\sigma \bar{v}_k^j$ and $\partial_\tau \bar{v}_k^j$ belong to the Hölder class $C^{0,\alpha}$ for all $0 < \alpha < 1$. Thus again by Arzelá-Ascoli theorem, $\partial_\sigma \bar{v}_k^j$ and $\partial_\tau \bar{v}_k^j$ admit subsequences which converge. Thus

$$\bar{v}_\infty^j := \beta_j v_\infty \in C_c^1(\mathbb{C}, \mathbb{R}^{2n})$$

for all $j \in \mathbb{N}$ with

$$v_k|_{B_{r_j}(z_j)} \xrightarrow{C^1} \bar{v}_\infty^j.$$

Hence we have showed that $v_\infty \in C^1(\mathbb{C}, M)$ and

$$v_k \xrightarrow{C^1_{\text{loc}}} v_\infty$$

modulo a subsequence. This procedure can be generalised to higher derivatives and is referred to as ***elliptic bootstrapping***. Thus we get $v_\infty \in C^\infty(\mathbb{C}, M)$ and

$$v_k \xrightarrow{C^\infty_{\text{loc}}} v_\infty$$

modulo a subsequence as $k \to \infty$. Choose $t_\infty \in \mathbb{S}^1$ such that $t_k' \to t_\infty$. Then v_∞ satisfies

$$\partial_\sigma v_\infty + J_{t_\infty} \partial_\tau v_\infty = 0.$$

Finally, v_∞ is nonconstant, as

$$\|\partial_\sigma v_\infty\|_{L^\infty} = \lim_{k\to\infty} \|\partial_\sigma v_k\|_{L^\infty} \geq 1$$

by Lemma 3.3. □

3.5 Removal of Singularity

So far we have not made use of the uniform bound on the energy $E(u_k)$. This will be crucial for extending the non-constant J_{t_∞}-holomorphic plane

$$v_\infty \colon \mathbb{C} \to M$$

from Lemma 3.5 to a non-constant J_{t_∞}-holomorphic sphere

$$v \colon \mathbb{S}^2 \to M, \qquad v|_{\mathbb{C}} = v_\infty.$$

In order to remove a singularity, we add a singularity at the origin of $\mathbb{C}$. For that we consider the conformal diffeomorphism

$$\varphi \colon \mathbb{C} \setminus \{0\} \to \mathbb{R} \times \mathbb{S}^1, \qquad \varphi(re^{i\theta}) := \log(re^{i\theta}), \tag{12}$$

and

$$w \colon \mathbb{R} \times \mathbb{S}^1 \to M, \qquad w := v_\infty \circ \varphi^{-1}.$$

Then

$$\lim_{s \to -\infty} w(s, \cdot) = v_\infty(0)$$

and w is of finite energy. Indeed, we have that

$$E(w) = \int_{\mathbb{R}\times\mathbb{S}^1} w^*\omega = \int_{-\infty}^{+\infty} \int_0^1 \omega(\partial_s w(s,t), J_{t_\infty} \partial_s w(s,t))\, dtds \leq C$$

by [MS12, Lemma 2.2.1]. The removal of singularities of v_∞ is now equivalent to show the existence of a positive constant asymptotic of w. Indeed, w is a negative gradient flow line of the symplectic area functional

$$\mathscr{A}_0 \colon \widetilde{\mathscr{L}_0 M} \to \mathbb{R}, \qquad \mathscr{A}_0([\gamma, \bar{\gamma}]) := -\int_{\mathbb{D}} \bar{\gamma}^*\omega,$$

where the Novikov covering space $\widetilde{\mathscr{L}_0 M}$ of $\mathscr{L}_0 M$ consists of all equivalence classes $[\gamma, \bar{\gamma}]$ of contractible loops $\gamma \in \mathscr{L}_0 M$ and fillings $\bar{\gamma}$ of γ such that

$$(\gamma, \bar{\gamma}) \sim (\gamma, \bar{\gamma}') \qquad :\Leftrightarrow \qquad \int_{\mathbb{D}} \bar{\gamma}^*\omega = \int_{\mathbb{D}} \bar{\gamma}'^*\omega \qquad \Leftrightarrow \qquad \omega([\bar{\gamma} \# (-\bar{\gamma}')]) = 0.$$

Note that $w(s, \cdot) \in \mathscr{L}_0 M$ by construction. As $\operatorname{Crit} \mathscr{A}_0 \cong M \times \pi_2(M) / \ker(\omega|_{\pi_2(M)})$, we have that w asymptotically converges to points in M. See Figure 7.

In order to do that, we consider the reparametrised sequence

$$w_k \colon \mathbb{R} \times \mathbb{S}^1 \to M, \qquad w_k(s,t) := w(s + r_k, t), \tag{13}$$

where $(r_k) \subseteq \mathbb{R}$ is a sequence with $r_k \to +\infty$ as $k \to \infty$. The aim of this section is to show that the derivatives of w_k are uniformly bounded. Consider the energy density $e \in C^\infty(\mathbb{R} \times \mathbb{S}^1, [0, +\infty))$ defined by

$$e(s,t) := \|\partial_s w(s,t)\|_{g_J}^2 .$$

where $J := J_{t_\infty}$.

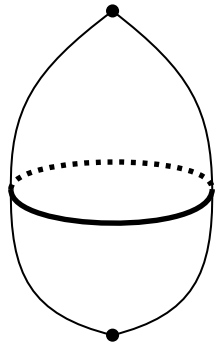


Figure 7: The J-holomorphic curve w as a negative gradient flow line of the symplectic area functional $\mathscr{A}_0$ with asymptotic ends .

Lemma 3.6. *There exists a constant $a \geq 0$ such that*

$$\Delta e \geq -ae^2. \tag{14}$$

Proof. Let R denote the curvature tensor of the Levi–Civita connection induced by the Riemannian metric $g_J(\cdot,\cdot) := \omega(\cdot, J\cdot)$ and abbreviate

$$\xi := w_*(\partial_s) = \partial_s w \qquad \text{and} \qquad \eta := w_*(\partial_t) = \partial_t w.$$

Then we compute

$$\begin{aligned}
\Delta e &= \partial_s^2 e + \partial_t^2 e \\
&= \partial_s \partial_s g_J(\xi,\xi) + \partial_t \partial_t g_J(\xi,\xi) \\
&= 2\partial_s g_J(\xi, \nabla_s \xi) + 2\partial_t g_J(\xi, \nabla_t \xi) \\
&= 2\left\|\nabla_s \xi\right\|_{g_J}^2 + 2\left\|\nabla_t \xi\right\|_{g_J}^2 + 2g_J(\xi, \nabla_s \nabla_s \xi + \nabla_t \nabla_t \xi) \\
&= 2\left\|\nabla_s \xi\right\|_{g_J}^2 + 2\left\|\nabla_t \xi\right\|_{g_J}^2 + 2g_J(\xi, \nabla_s \nabla_s \xi + \nabla_t \nabla_t \xi) \\
&= 2\left\|\nabla_s \xi\right\|_{g_J}^2 + 2\left\|\nabla_t \xi\right\|_{g_J}^2 + 2g_J(\xi, \nabla_s \nabla_s \xi + \nabla_s \nabla_t \eta - \nabla_s \nabla_t \eta + \nabla_t \nabla_t \xi) \\
&= 2\left\|\nabla_s \xi\right\|_{g_J}^2 + 2\left\|\nabla_t \xi\right\|_{g_J}^2 + 2g_J(\xi, \nabla_s (\nabla_s \xi + \nabla_t \eta) - \nabla_s \nabla_t \eta + \nabla_t \nabla_s \eta) \\
&= 2\left\|\nabla_s \xi\right\|_{g_J}^2 + 2\left\|\nabla_t \xi\right\|_{g_J}^2 + 2g_J(\xi, \nabla_s (\nabla_s \xi + \nabla_t \eta) - R(\xi,\eta)\eta) \\
&= 2\left\|\nabla_s \xi\right\|_{g_J}^2 + 2\left\|\nabla_t \xi\right\|_{g_J}^2 + 2g_J(\xi, \nabla_s (\nabla_t (J\xi) - \nabla_s (J\eta)) - R(\xi,\eta)\eta) \\
&= 2\left\|\nabla_s \xi\right\|_{g_J}^2 + 2\left\|\nabla_t \xi\right\|_{g_J}^2 + 2g_J(\xi, \nabla_s ((\nabla_\eta J)\xi - (\nabla_\xi J)\eta) - R(\xi,\eta)\eta)
\end{aligned}$$

by [Lee18, Lemma 6.2] and [Lee18, Proposition 7.5]. Thus

$$\frac{1}{2}\Delta e = \left\|\nabla_s \xi\right\|_{g_J}^2 + \left\|\nabla_t \xi\right\|_{g_J}^2 - g_J(\xi, R(\xi,\eta)\eta) + \kappa,$$

with

$$\kappa := g_J(\xi, \nabla_s ((\nabla_\eta J)\xi - (\nabla_\xi J)\eta))$$

$$= g_J(\xi, (\nabla_s(\nabla_\eta J))\xi - (\nabla_s(\nabla_\xi J))\eta) + g_J(\xi, (\nabla_\eta J)\nabla_s\xi - (\nabla_\xi J)\nabla_s\eta).$$

Now we estimate κ. We have that

$$\|\nabla_s(\nabla_\eta J)\|_{L^\infty} \le C_0(\|\eta\|_{g_J} \|\xi\|_{g_J} + \|\nabla_s\eta\|_{g_J}) = C_0(\|\xi\|_{g_J}^2 + \|\nabla_t\xi\|_{g_J}),$$

and similarly

$$\|\nabla_t(\nabla_\xi J)\|_{L^\infty} \le C_1(\|\xi\|_{g_J}^2 + \|\nabla_s\xi\|_{g_J}).$$

Consequently

$$\kappa \ge -2C_2(\|\xi\|_{g_J}^4 + \|\xi\|_{g_J}^2 (\|\nabla_s\xi\|_{g_J} + \|\nabla_t\xi\|_{g_J})).$$

Let $\varepsilon > 0$. Then

$$\begin{aligned}\frac{1}{2}\Delta e &\ge \|\nabla_s\xi\|_{g_J}^2 + \|\nabla_t\xi\|_{g_J}^2 - C_3\|\xi\|_{g_J}^4 - \frac{C_3}{2\varepsilon^2}\|\xi\|_{g_J}^4 - \frac{C_3\varepsilon^2}{2}(\|\nabla_s\xi\|_{g_J} + \|\nabla_t\xi\|_{g_J})^2 \\ &\ge \|\nabla_s\xi\|_{g_J}^2 + \|\nabla_t\xi\|_{g_J}^2 - C_3\|\xi\|_{g_J}^4 - \frac{C_3}{2\varepsilon^2}\|\xi\|_{g_J}^4 - C_3\varepsilon^2(\|\nabla_s\xi\|_{g_J}^2 + \|\nabla_t\xi\|)_{g_J}^2.\end{aligned}$$

Choosing $\varepsilon := C_3^{-1/2}$ finally yields

$$\frac{1}{2}\Delta e \ge -ae^2, \qquad a := C_3 + \frac{C_3^2}{2}.$$

□

Lemma 3.7 (A priori estimate). *For some $r > 0$ and $a \ge 0$, suppose there is a C^2 function $e : B_r(0) \to [0, \infty)$ satisfying*

$$\Delta e \ge -ae^2, \qquad \int_{B_r(0)} e\, dsdt < \frac{\pi}{8a}. \tag{15}$$

Then we have

$$e(0) \le \frac{8}{\pi r^2}\int_{B_r(0)} e\, dsdt. \tag{16}$$

Proof. We first show some statments in the following several steps.

Step 1. Show that if there is some $b \ge 0$ s.t. a function $e \in C^2(B_r(0), [0, \infty))$ satisfying $\Delta e \ge -b$, then

$$e(0) \le \frac{br^2}{8} + \frac{1}{\pi r^2}\int_{B_r(0)} e\, dsdt.$$

We define a function

$$f(s,t) = e(s,t) + \frac{b}{4}(s^2 + t^2),$$

and by assumption $\Delta f = \Delta e + b \geq 0$, i.e. f is subharmonic. Due to the mean value inequality for subharmonic function (it follows from divergence theorem, see the proof of [MS12, Lemma 4.3.3]):

$$f(0) \leq \frac{1}{\pi r^2} \int_{B_r(0)} f \, dsdt,$$

we can compute that

$$\begin{aligned} e(0) = f(0) &\leq \frac{1}{\pi r^2} \int_{B_r(0)} f \, dsdt \\ &\overset{\substack{z=s+it \\ =\rho e^{i\theta}}}{=} \int_0^r \int_0^{2\pi} f(\rho e^{i\theta}) \rho d\theta d\rho \\ &= \frac{1}{\pi r^2} \int_{B_r(0)} e \, dsdt + \frac{1}{\pi r^2} 2\pi \frac{b}{4} \int_0^r \rho^3 \, d\rho \\ &= \frac{1}{\pi r^2} \int_{B_r(0)} e \, dsdt + \frac{br^2}{8}. \end{aligned}$$

This finishes **Step 1**.

Step 2. Show that w.l.o.g. we can put $r = 1$. We assume function $e \in C^2(B_r(0), [0, \infty))$ satisfies condition (15) for some r and a. We rescale the function e to obtain another function $\tilde{e} \in C^2(B_1(0), [0, \infty))$ by $\tilde{e}(s, t) := e(rs, rt)$. Now $\tilde{e}$ satisfies (15) for $\tilde{a} := r^2 a$: This can be seen by computations

$$\Delta \tilde{e}(s, t) = r^2 \Delta e(rs, rt) \geq -ar^2 e(rs, rt) = -\tilde{a}\tilde{e}(s, t)^2$$

and

$$\int_{B_1(0)} \tilde{e} \, dsdt = \frac{1}{r^2} \int_{B_r(0)} e \, dsdt \leq \frac{\pi}{8ar^2} = \frac{\pi}{8\tilde{a}}.$$

Next, we assume the statement of lemma holds for the case $r = 1$, then

$$e(0) = \tilde{e}(0) \leq \frac{8}{\pi} \int_{B_1(0)} \tilde{e} \, dsdt = \frac{8}{\pi r^2} \int_{B_r(0)} e \, dsdt.$$

This finishes **Step 2**.

Step 3. Show that w.l.o.g. we can put $a = 1$. The trivial case is when $a = 0$, then e is subharmonic function, and we are done for the lemma. When $a > 0$, we put $\tilde{e} := ae$, then $\tilde{e}$ satisfies (15) for $a = 1$. Now assume the statement of the lemma holds for $a = 1$, then we have

$$e(0) = \frac{1}{a}\tilde{e}(0) \leq \frac{8}{a\pi r^2} \int_{B_r(0)} \tilde{e} \, dsdt = \frac{8}{\pi r^2} \int_{B_r(0)} e \, dsdt.$$

This finishes **Step 3**.

Step 4 (Heinz trick). Suppose $e \in C^2(B_1(0), [0,\infty))$ satisfies (15) for $r = 1, a = 1$, and define a function $\psi : [0,1] \to [0,\infty)$ by $\psi(\rho) := (1-\rho)^2 \sup_{B_\rho(0)} e$. Pick $\rho^* \in [0,1)$ s.t. $\psi(\rho^*) = \max_{\rho\in[0,1]} \psi(\rho)$, and put $c := \sup_{B_{\rho^*}(0)} e$. Then we can show that for $\rho \in [0, \frac{1-\rho^*}{2}]$,

$$c \le 2c^2\rho^2 + \frac{1}{\pi\rho^2}\int_{B_1(0)} e\, dsdt.$$

First notice that $B_{\rho^*}(0)$ as a closed ball is compact set $\Rightarrow \exists\, z^* \in B_{\rho^*}(0)$ s.t. $e(z^*) = \sup_{B_{\rho^*}(0)} e =: c$. Put $\varepsilon := \frac{1-\rho^*}{2}$, and we find

$$\begin{aligned}\sup_{B_\varepsilon(z^*)} e &\le \sup_{B_{\rho^*+c}(0)} e =: \frac{\psi(\rho^*+\varepsilon)}{(1-\rho^*-\varepsilon)^2} = \frac{4\psi(\rho^*+\varepsilon)}{(1-\rho^*)^2} \\ &\le \frac{4\psi(\rho^*)}{(1-\rho^*)^2} = 4\sup_{B_{\rho^*}(0)} e =: 4c\end{aligned}$$

$\Rightarrow$

$$\Delta e \ge -e^2 \ge -16c^2 \qquad \text{on } B_\varepsilon(z^*).$$

Together with **Step 1**, this leads to the following on $B_\rho(z^*) \subseteq B_\varepsilon(z^*)$ (i.e. $\rho \le \varepsilon$):

$$\begin{aligned}c = e(z^*) &\le \frac{16c^2\rho^2}{8} + \frac{1}{\pi\rho^2}\int_{B_\rho(z^*)} e\, dsdt \\ &\le 2c^2\rho^2 + \frac{1}{\pi\rho^2}\int_{B_1(0)} e\, dsdt.\end{aligned}$$

This finishes **Step 4**.

We are now ready to prove the lemma: By **Step 2** and **Step 3**, w.l.o.g. we put $r = 1, a = 1$, and c, ε as in **Step 4**.

Claim. $4c\varepsilon^2 < 1$.

We prove the claim by contradiction: Suppose not, and apply **Step 4** to $\rho = \sqrt{\frac{1}{4c}} \le \varepsilon$. We have

$$c \le \frac{c}{2} + \frac{4c}{\pi}\int_{B_1(0)} e\, dsdt \Rightarrow \int_{B_1(0)} e\, dsdt \ge \frac{\pi}{8}.$$

This contradicts to the condition (15).

Put $\rho = \varepsilon$ in **Step 4**, and we get

$$\begin{aligned}c &\le 2c^2\varepsilon^2 + \frac{1}{\pi\varepsilon^2}\int_{B_1(0)} e\, dsdt \\ &\overset{\textbf{Claim}}{\le} \frac{c}{2} + \frac{1}{\pi\varepsilon^2}\int_{B_1(0)} e\, dsdt\end{aligned}$$

$\Rightarrow$

$$c \leq \frac{2}{\pi\varepsilon^2} \int_{B_1(0)} e \, dsdt$$

$\Rightarrow$

$$\begin{aligned} e(0) = \psi(0) \leq \psi(\rho^*) = (1-\rho^*)^2 c = 4\varepsilon^2 c \\ \leq \frac{8}{\pi} \int_{B_1(0)} e \, dsdt. \end{aligned}$$

This concludes the proof of the lemma. □

Lemma 3.8. *There exists $w_\infty \in M$ such that*

$$w_k \xrightarrow{C^\infty_{\text{loc}}} w_\infty, \qquad k \to \infty$$

passing through a subsequence.

Proof. Let $a \geq 0$ denote the constant from Lemma 3.6. As $E(w) < +\infty$, there exists $R > 0$ such that

$$\int_R^{+\infty} \int_0^1 e(s,t) \, dtds \leq \min\left\{\frac{\pi}{8a}, 1\right\}$$

and

$$\int_{-\infty}^{-R} \int_0^1 e(s,t) \, dtds \leq \min\left\{\frac{\pi}{8a}, 1\right\}.$$

Let $z = (s,t) \in \mathbb{R} \times \mathbb{S}^1$ such that $|s| \geq R+1$. Then

$$\int_{B_1(z)} e \, dsdt < \frac{\pi}{8a}.$$

By Lemma 3.7 we conclude

$$e(z) \leq \frac{8}{\pi} \int_{B_1(z)} e \, dsdt \leq \frac{8}{\pi}.$$

Consequently, e is uniformly bounded, as $e|_{[-R-1,R+1]\times\mathbb{S}^1}$ is uniformly bounded by continuity of e. This implies that $\|\partial_s w\|_{g_J}$ is uniformly bounded by definition of e. As w is a J-holomorphic curve, we also conclude that $\|\partial_t w\|_{g_J}$ is uniformly bounded. Hence also the derivatives of w_k are uniformly bounded. Thus by an eliptic bootrstrapping argument as in Subsection 3.4, we conclude

$$w_k \xrightarrow{C^\infty_{\text{loc}}} w_\infty, \qquad k \to \infty$$

up to a subsequence, where w_∞ is a negative gradient flow line of the area functional $\mathscr{A}_0$. We show that w_∞ is constant. Towards a contradiction, assume that w_∞ is not constant. Thus there exists $s < s'$ such that

$$\varepsilon := \mathscr{A}_0(w_\infty(s,\cdot)) - \mathscr{A}_0(w_\infty(s',\cdot)) > 0.$$

Moreover, there exists $N \in \mathbb{N}$ such that

$$\mathscr{A}_0(w_k(s,\cdot)) - \mathscr{A}_0(w_k(s',\cdot)) = \mathscr{A}_0(w(s+r_k,\cdot)) - \mathscr{A}_0(w(s'+r_k,\cdot)) \geq \frac{\varepsilon}{2}$$

for all $k \geq K$. Define a subsequence (r_{k_j}) of (r_k) recursively by

$$k_0 := N \qquad \text{and} \qquad k_j := \min\{\ell \in \mathbb{N} : s' + r_{k_{j-1}} \leq s + r_\ell\}.$$

This works as $r_k \to +\infty$ as $k \to \infty$. Fix $\ell \in \mathbb{N}$. Then we compute

$$\begin{aligned} E(w) &= \sup_{s\in\mathbb{R}} \mathscr{A}_0(w(s,\cdot)) - \inf_{s\in\mathbb{R}} \mathscr{A}_0(w(s,\cdot)) \\ &\geq \mathscr{A}_0(w(s+r_{k_0},\cdot)) - \mathscr{A}_0(w(s+r_{k_\ell},\cdot)) \\ &= \sum_{\nu=1}^{\ell} \big(\mathscr{A}_0(w(s+r_{k_{\nu-1}},\cdot)) - \mathscr{A}_0(w(s+r_{k_\nu},\cdot))\big) \\ &\geq \sum_{\nu=1}^{\ell} \big(\mathscr{A}_0(w(s+r_{k_{\nu-1}},\cdot)) - \mathscr{A}_0(w(s'+r_{k_{\nu-1}},\cdot))\big) \\ &\geq \frac{\ell\varepsilon}{2}, \end{aligned}$$

as $\mathscr{A}_0$ is a Lyapunov function for the negative gradient flow of $\operatorname{grad}_J \mathscr{A}_0$. This contradicts the finiteness of $E(w)$. □

Remark 3.5. A priori, w_∞ might depend on the choice of subsequence (r_{k_j}). If that was the case, then w_∞ would not be unique from the procedure of extracting convergent subsequences. In general, one can think of some bad examples for gradient-like system, e.g. having ω-limit set as limit cycle, and extracting different subsequences along the flow line would converge to different points on the limit cycle.

The way to resolve the uniqueness issue for w_∞ is to regard the area functional $\mathscr{A}_0$ roughly as "Morse" function, then we can reveal the exponential decay behavior of its gradient flow line converging towards a critical point, which is referred to w_∞ here, namely we actually have the following result: For sufficiently large $s_0 > 0$, there exists constants $c > 0, \delta > 0$ s.t.

$$d_{L^2}(w(s,\cdot), w_\infty) \leq ce^{-\delta s}, \qquad \forall\, s \geq s_0 \tag{17}$$

where d_{L^2} is the L^2-distance on $\mathscr{L}_0 M \subset C^\infty(\mathbb{S}^1, M)$ w.r.t. L^2-inner product. The detail discussion on exponential decay will be left to the next section.

Lemma 3.9 (Removal of singularity). *For the gradient flow line w of* $\operatorname{grad}_J \mathscr{A}_0$ *of finite energy, i.e. $E(w) < \infty$, $\exists!\, w_\infty \in M$ s.t.*

$$w(s,\cdot) \xrightarrow{C^\infty_{loc}} w_\infty \qquad \text{as } s \to +\infty.$$

Proof. The inequality (17) implies

$$w(s,\cdot) \xrightarrow{L^2_{\mathrm{loc}}} w_\infty, \qquad s \to +\infty.$$

Automatically, this leads to

$$w(s,\cdot) \xrightarrow{C^\infty_{\mathrm{loc}}} w_\infty, \qquad s \to +\infty.$$

We can see this by contradiction: Assume that the C^∞_{loc}-convergence does not hold. Then there exists a C^∞-open neighbourhood $\mathscr{U}$ of w_∞ in $\mathscr{L}_0 M$ and a sequence $(r_k) \subseteq (0,+\infty)$ s.t. $\lim_{k\to\infty} r_k = +\infty$ and $w(r_k,\cdot) \notin \mathscr{U}$ $\forall\, k \in \mathbb{N}$. By Lemma 3.8, we have that $w(r_k)$ converges up to subsequences. But we know

$$w(s,\cdot) \xrightarrow{L^2_{\mathrm{loc}}} w_\infty, \qquad s \to +\infty,$$

this enforces

$$w(r_k,\cdot) \xrightarrow{C^\infty_{\mathrm{loc}}} w_\infty, \qquad s \to +\infty,$$

contradicting $w(r_k,\cdot) \notin \mathscr{U}$.

For uniqueness of w_∞, suppose w_∞ and w'_∞ are obtained from subsequences (r_k) and (r'_k), by (17) we have

$$\begin{aligned} d_{L^2}(w_\infty, w'_\infty) &\le d_{L^2}(w(s,\cdot), w_\infty) + d_{L^2}(w(s,\cdot), w'_\infty) \\ &\le ce^{-\delta s} + c'e^{-\delta' s} \xrightarrow{s\to+\infty} 0. \end{aligned}$$

In this way, we see that extracting convergent subsequences can only lead to unique result. With the existence and uniqueness of w_∞, we completes the proof of removal of singularity. □

Combining the results achieved so far, we finally obtain a non-constant pseudoholomorphic sphere, i.e. a "bubble", on the Floer cylinder after passing through converging subseqeunce. Hence, we have proved Theorem 3.4.

3.6 Exponential Decay

In this section, we aim to discuss the exponential convergence of the flow line of $\mathrm{grad}_J \mathscr{A}_0$. Before we start to deal with the main business, we shall first recall briefly the picture from finite dimensional Morse theory: Around each critical point of a Morse function $f : M \to \mathbb{R}$, there is a local Morse coordinate chart $\varphi : M \supset U \to V \subset \mathbb{R}^n$ centered at the critical point, in which the Morse function is expressed as a quadratic function in terms of Morse coordinate $x = (x_1, \ldots, x_n)$:

$$f(x) = f(0) + \frac{1}{2}\langle x, Ax\rangle$$

where A is given by the Hessian of the Morse function at the critical point

$$A = \operatorname{Hess} f = \begin{pmatrix} a_1 & & \\ & \ddots & \\ & & a_n \end{pmatrix}.$$

It is a general strategy to study non-linear system by linearizing it locally, so we would like to observe the gradient flow equation on Riemannian manifold (M, g), which is a non-linear ODE

$$\dot{x}(s) + \operatorname{grad}_g f(x(s)) = 0, \qquad x \in C^\infty(\mathbb{R}, M),$$

in the Morse chart (U, φ) w.r.t. standard Euclidean metric $g_0(\cdot, \cdot) = \langle \cdot, \cdot \rangle$ on $V \subset \mathbb{R}^n$. This requires an assumption that the chosen Riemannian metric g on the manifold M is locally flat restricting on that Morse chart, i.e. $g|_U = \varphi^* g_0|_V$. Note that this certainly can not be true the general, since φ is not isometry, and when there is non-zero curvature, such assumption fails. However, we can choose $U \Subset M$ small enough so that locally we have $\frac{1}{\kappa^2} g \leq g_0 \leq \kappa^2 g$ for some constant $\kappa > 0$. This enables us to first analyse the dynamical behavior in the flat model on $V \subset \mathbb{R}^n$ and then pullback the information on $U \subset M$ via the technique of action-energy inequality. The advantage of flat model is that we have decent explicit description on the linear system: The gradient flow equation in such a flat model yields the form of hyperbolic linear system:

$$\dot{x}(s) + Ax(s) = 0, \qquad x \in C^\infty(\mathbb{R}, \mathbb{R}^n)$$

thus it has exponential convergence behavior (c.f. [Tes12, Theorem 9.2]). This can be seen explicitly: The Hessian matrix is diagonalized into the diagonal matrix with diagonal entries as the eigenvalues, then restricting to positive resp. negative eigenvalues, the corresponding coordinate components in the solution of gradient flow equation will have exponential decay to 0 as $t \to +\infty$ resp. $t \to -\infty$, i.e. the origin of the Morse chart. Eventually, for $x \in C^\infty(\mathbb{R}, M)$ a gradient flow line with asymptotic convergence to critical point $x_\infty = x(+\infty)$, there exist constants $c > 0, \delta > 0$ and some $s_0 \in \mathbb{R}$ s.t.

$$d_g(x(s), x_\infty) \leq c e^{-\delta s}, \qquad \forall\, s \geq s_0.$$

More detail can be found in [FN20, §6].

The action functional $\mathscr{A}_H : \mathscr{L}_0 M \to \mathbb{R}$ for non-degenerate Hamiltonian H on symplectically aspherical manifold, as mentioned at the very beginning in the introduction, could be regarded as a Morse function on the infinite dimensional manifold. However, we should work in a slightly more general setting called the Morse–Bott theory, namely the action functional $\mathscr{A}_H : \widetilde{\mathscr{L}_0 M} \to \mathbb{R}$ for a broader class of Hamiltonians on general symplectic manifold could be regarded as a Morse–Bott function. We give a brief description on Morse–Bott function and refer the reader to the appendix A for more detail: As a mild generalization to

Morse function, a Morse–Bott function $f : M \to \mathbb{R}$ allows certain degeneracy, namely its critical points form a continuum termed critical submanifold $C \subseteq M$ consisting of disjoint connected components. The critical submanifold satisfies the conditions: $\ker \operatorname{Hess} f(x) = T_x C$ and $\operatorname{Hess} f(x)|_{\nu_x C}$ is non-degenerate $\forall\, x \in C$, the cirtical submanifold is also called the Morse–Bott submanifold. The normal form of the Hessian matrix A is similar to that of Morse function, despite that it now contains zero diagonal entries corresponding to the tangential directions on critical manifold. There is analogy to Morse lemma called Morse–Bott lemma, which states the existence of local Morse–Bott coordinate and the quadratic expression of Morse–Bott function with the Hessian in the Morse–Bott chart. Thus, we can similarly consider the flat model as in Morse case, in addition to the alike behavior from the positive/negative eigenvalues, the zero eigenvalues imply the corresponding coordinate components stay constant. The reason we take the Morse–Bott setting into account is that in the special case $H = 0$, the action functional becomes the so-called area functional $\mathscr{A}_0$, which is clearly degenerate since it has $M \subset \widetilde{\mathscr{L}_0 M}$ as critical manifold. Principally, a Hamiltonian (autonomous or time-dependent) are said to be of Morse–Bott type if its 1-periodic orbit set degenerate into Morse–Bott submanifold, e.g. $H = 0$ whose periodic orbits are points forming disjoint copies of M. In fact, Morse–Bott Hamiltonian gives rise Morse–Bott action functional.

To construct a flat model for the infinite dimensional Morse–Bott function

$$\mathscr{A}_0(\gamma) = -\int_{\mathbb{D}} \overline{\gamma}^* \omega, \qquad \gamma \in C^\infty(\mathbb{S}^1, M),$$

we apply Darboux's theorem and choose a Darboux chart so that we work in standard symplectic space

$$(\mathbb{C}^n, \frac{i}{2}\sum_{i=1}^n dz_i \wedge d\bar{z}_i) \cong (\mathbb{R}^{2n}, \sum_{i=1}^n dx_i \wedge dy_i, J_0)$$

where $J_0 = i$ is the standard complex structure. Here we also implicitly assume that the almost complex J structure on M is locally linear in Darboux chart and coincides with the standard complex structure i on $\mathbb{C}^n$, which would not be true in general. Now we pick a loop $z \in C^\infty(\mathbb{S}^1, \mathbb{C}^n)$ and perform Fourier expansion[1]:

$$z(t) = \sum_{k=-\infty}^{+\infty} z_k e^{2\pi i k t}, \qquad z_k \in \mathbb{C}^n, k \in \mathbb{Z}.$$

We write $z = (z^1, \ldots, z^n)$ with $z^j \in C^\infty(\mathbb{S}^1, \mathbb{C})$ for all $j = 1, \ldots, n$. The Fourier expansion yields:

$$z^j(t) = \sum_{k=-\infty}^{+\infty} z_k^j e^{2\pi i k t}, \qquad z_k^j = x_k^j + i y_k^j, \qquad x_k^j, y_k^j \in \mathbb{R}.$$

[1] Recall that the family $(e^{2\pi i k t})_{k\in\mathbb{Z}}$ form a unitary basis of $C^0(\mathbb{S}^1, \mathbb{C}^n)$ by Stone-Weierstrass theorem since $\mathbb{S}^1$ is compact, and the functions in $C^0(\mathbb{S}^1, \mathbb{C}^n)$ are bounded, hence $C^\infty(\mathbb{S}^1, \mathbb{C}^n)$ is dense in $C^0(\mathbb{S}^1, \mathbb{C}^n)$.

Lemma 3.10. *Let z be as described above. Then*

$$\mathscr{A}_0(z) = \frac{1}{2}\int_0^1 iz(t)\cdot\dot{z}(t)\,dt = -\pi\sum_{-\infty}^{+\infty} k\sum_{j=1}^{n}|z_k^j|^2 = -\pi\sum_{k=-\infty}^{+\infty} k\sum_{j=1}^{n}[(x_k^j)^2+(y_k^j)^2].$$

Proof. We denote the right hand side as

$$\mathbf{A}(z) := -\pi\sum_{-\infty}^{+\infty} k\sum_{j=1}^{n}|z_k^j|^2 = -\pi\sum_{k=-\infty}^{+\infty} k\sum_{j=1}^{n}[(x_k^j)^2+(y_k^j)^2]$$

Clearly, we have $\mathscr{A}_0(0) = \mathbf{A}(0)$. Since $C^\infty(\mathbb{S}^1,\mathbb{C}^n)$ is connected, it suffices to show:

$$d\mathscr{A}_0(z) = d\mathbf{A}(z), \qquad \forall\, z\in C^\infty(\mathbb{S}^1,\mathbb{C}^n).$$

Let $\xi\in T_zC^\infty(\mathbb{S}^1,\mathbb{C}^n)\cong\Gamma(z^*T\mathbb{C}^n) = C^\infty(\mathbb{S}^1,\mathbb{C}^n)$, and write $\xi=(\xi^1,\dots,\xi^n)$. Under Fourier expansion:

$$\xi^j(t) = \sum_{k=-\infty}^{+\infty}\xi_k^j e^{2\pi ikt}, \qquad \xi_k^j\in\mathbb{C}.$$

Then we compute the Gâteaux differential:

$$\begin{aligned}
d\mathscr{A}_0(z)\xi &= \int_0^1 \underbrace{\omega(\partial_t z(t),\xi(t))}_{-\operatorname{Re}\sum_{j=1}^n i\overline{\partial_t z^j(t)}\xi^j(t)}\,dt\\
&= -\operatorname{Re}\sum_{j=1}^{n}\int_0^1 \underbrace{i\overline{\left(\sum_{k=-\infty}^{+\infty}2\pi ikz_k^j e^{2\pi ikt}\right)}}_{-2\pi\sum_{k=-\infty}^{+\infty}ik\bar{z}_k^j e^{-2\pi ikt}}\left(\sum_{l=-\infty}^{+\infty}\xi_l^j e^{2\pi ilt}\right)dt\\
&= -2\pi\operatorname{Re}\sum_{j=1}^{n}\sum_{k=-\infty}^{+\infty}\sum_{l=-\infty}^{+\infty}\int_0^1 k\bar{z}_k^j\xi_l^j e^{2\pi i(l-k)t}dt\\
&= -2\pi\operatorname{Re}\sum_{j=1}^{n}\sum_{k=-\infty}^{+\infty}\sum_{l=-\infty}^{+\infty}k\bar{z}_k^j\xi_k^j\\
&= d\mathbf{A}(z)\xi.
\end{aligned}$$

This finishes the proof. □

We can think of $(\dots,z_{k-1},z_k,z_{k+1},\dots)$ as infinite dimensional Morse–Bott coordinate of z. Then the lemma plays the role of Morse–Bott lemma for $\mathscr{A}_0: C^\infty(\mathbb{S}^1,\mathbb{C}^n)\to\mathbb{R}$. Choosing some suitable inner product $\langle\cdot,\cdot\rangle$ on $C^\infty(\mathbb{S}^1,\mathbb{C}^n)$, we have

$$\mathscr{A}_0(z) = \frac{1}{2}\langle z, Az\rangle$$

where z on right hand side represents the infinite dimensional Morse–Bott coordinate column vector mentioned above, and $A = \operatorname{Hess} \mathscr{A}_0(z)$. The next step is to analyse the Hessian and its eigenvalues w.r.t. a given inner product. Note that unlike the finite dimensional flat model, where all the metrics on finite dimensional vector space are equivalent, the infinite dimensional vector space has nonequivalent metrics, so the choice of inner product makes significant influence.

We discuss explicitly three cases below: Let $z \in C^\infty(\mathbb{S}^1, \mathbb{C}^n)$, and $\xi, \eta \in T_z C^\infty(\mathbb{S}^1, \mathbb{C}^n) \cong C^\infty(\mathbb{S}^1, \mathbb{C}^n)$. Under Fourier expansion, we have

$$\xi = (\xi^1, \dots, \xi^j, \dots, \xi^n), \qquad \xi^j(t) = \sum_{k=-\infty}^{+\infty} \xi_k^j e^{2\pi i k t}$$

$$\eta = (\eta^1, \dots, \eta^j, \dots, \eta^n), \qquad \eta^j(t) = \sum_{k=-\infty}^{+\infty} \eta_k^j e^{2\pi i k t}.$$

• L^2-inner product:

$$\begin{aligned}
\langle \xi, \eta \rangle_{L^2} &= \int_0^1 \operatorname{Re} \sum_{k=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} \bar{\xi}_k \cdot \eta_l e^{2\pi i (l-k) t} \, dt \\
&= \sum_{k=-\infty}^{+\infty} \sum_{j=1}^{n} \operatorname{Re}(\bar{\xi}_k^j \eta_k^j) \\
&= \sum_{k=-\infty}^{+\infty} \sum_{j=1}^{n} (\operatorname{Re} \xi_k^j \operatorname{Re} \eta_k^j + \operatorname{Im} \xi_k^j \operatorname{Im} \eta_k^j).
\end{aligned}$$

The action functional reads:

$$\mathscr{A}_0(z) = -\pi \sum_{k=-\infty}^{+\infty} k \sum_{j=1}^{n} |z_k^j|^2 = -\pi \sum_{k=-\infty}^{+\infty} k \sum_{j=1}^{n} [(x_k^j)^2 + (y_k^j)^2].$$

Under the Morse–Bott coordinate $(\cdots, x_k^1, y_k^1, \cdots, x_k^n, y_k^n, \cdots)$, we write:

$$\operatorname{grad}_{L^2} \mathscr{A}_0(z) = \begin{pmatrix} \vdots \\ -2\pi k x_k^1 \\ -2\pi k y_k^1 \\ \vdots \\ -2\pi k x_k^n \\ -2\pi k y_k^n \\ \vdots \end{pmatrix},$$

$$\operatorname{Hess}_{L^2} \mathscr{A}_0(z) = \begin{pmatrix} \ddots & & & & & & & \\ & -2\pi k & & & & & & \\ & & -2\pi k & & & & & \\ & & & \ddots & & & & \\ & & & & -2\pi k & & & \\ & & & & & -2\pi k & & \\ & & & & & & \ddots & \end{pmatrix}.$$

The eigenvalues of the Hessian are $2\pi k$ ($k \in \mathbb{Z}$) with multiplicity $2n$, hence they are bounded from neither above nor below. However, there is a "gap" in the spectrum, i.e. 0 as a distinguished value is isolated in the spectrum. This implies the gradient flow line $z_s(t)$ has exponential decay with the rate of at least 2π when coverging to origin in the Morse–Bott coordinate. Consider the flow line converging as $s \to +\infty$, then starting at $z_{s_0} \in C^\infty(\mathbb{S}^1, \mathbb{C}^n)$ for some $s_0 \in \mathbb{R}$, we have:

$$d_{L^2}(z_s(\cdot), 0) = \|z_s(\cdot)\|_{L^2} \leq e^{-2\pi(s-s_0)} \|z_{s_0}(\cdot)\|_{L^2}, \qquad \forall\, s \geq s \geq s_0.$$

• $W^{1,2}$-inner product:

$$\begin{aligned} \langle \xi, \eta \rangle_{W^{1,2}} &= \int_0^1 \langle \xi, \eta \rangle \, dt + \int_0^1 \langle \partial_t \xi, \partial_t \eta \rangle \, dt \\ &= \sum_{k=-\infty}^{+\infty} \sum_{j=1}^{n} (\operatorname{Re} \xi_k^j \operatorname{Re} \eta_k^j + \operatorname{Im} \xi_k^j \operatorname{Im} \eta_k^j) \\ &\quad + 4\pi^2 \sum_{k=-\infty}^{+\infty} \sum_{j=1}^{n} k^2 (\operatorname{Re} \xi_k^j \operatorname{Re} \eta_k^j + \operatorname{Im} \xi_k^j \operatorname{Im} \eta_k^j) \end{aligned}$$

Under the Morse–Bott coordinate $(\cdots, x_k^1, y_k^1, \cdots, x_k^n, y_k^n, \cdots)$, we write:

$$\operatorname{grad}_{W^{1,2}} \mathscr{A}_0(z) = \begin{pmatrix} \vdots \\ -\frac{2\pi k}{1+4\pi^2k^2} x_k^1 \\ -\frac{2\pi k}{1+4\pi^2k^2} y_k^1 \\ \vdots \\ -\frac{2\pi k}{1+4\pi^2k^2} x_k^n \\ -\frac{2\pi k}{1+4\pi^2k^2} y_k^n \\ \vdots \end{pmatrix},$$

$$\operatorname{Hess}_{W^{1,2}} \mathscr{A}_0(z) = \begin{pmatrix} \ddots & & & & & & & \\ & -\frac{2\pi k}{1+4\pi^2k^2} & & & & & & \\ & & -\frac{2\pi k}{1+4\pi^2k^2} & & & & & \\ & & & \ddots & & & & \\ & & & & -\frac{2\pi k}{1+4\pi^2k^2} & & & \\ & & & & & -\frac{2\pi k}{1+4\pi^2k^2} & & \\ & & & & & & \ddots & \end{pmatrix}.$$

The eigenvalues of the Hessian are $\frac{2\pi k}{1+4\pi^2k^2}$ $(k \in \mathbb{Z})$ each with $2n$ multiplicity, and clearly the spectrum is bounded from above and below (comparing to L^2-inner product where the spectrum is not bounded). However

$$\frac{2\pi k}{1+4\pi^2k^2} \xrightarrow{k\to\infty} 0,$$

there is no "gap" in the spectrum, thus we do not have the exponential decay:

$$d_{W^{1,2}}(z_s(\cdot), 0) = \|z_s(\cdot)\|_{W^{1,2}} = \sum_{k\in\mathbb{Z}} e^{-\frac{2\pi k}{1+4\pi^2k^2}s}(1+2\pi|k|)|z_k^j|^2.$$

Note that the sum $\sum_{k\in\mathbb{Z}} e^{-\frac{2\pi k}{1+4\pi^2k^2}}$ alone diverges.

• $W^{\frac{1}{2},2}$-inner product: For $s \geq 0$, the fractional Sobolev inner product is defined as (here we concern only $s = \frac{1}{2}$):

$$\langle \xi, \eta \rangle_{W^{s,2}} := \langle \xi_0, \eta_0 \rangle + 2\pi \sum_{k\in\mathbb{Z}^\times} |k|^{2s} \langle \xi_k, \eta_k \rangle.$$

Under the Morse–Bott coordinate $(\cdots, x_k^1, y_k^1, \cdots, x_k^n, y_k^n, \cdots)$, we write:

$$\operatorname{grad}_{W^{\frac{1}{2},2}} \mathscr{A}_0(z) = \begin{pmatrix} \vdots \\ -\frac{2\pi k}{2\pi|k|} x_k^1 \\ -\frac{2\pi k}{2\pi|k|} y_k^1 \\ \vdots \\ -\frac{2\pi k}{2\pi|k|} x_k^n \\ -\frac{2\pi k}{2\pi|k|} y_k^n \\ \vdots \end{pmatrix},$$

$$\operatorname{Hess}_{W^{\frac{1}{2},2}} \mathscr{A}_0(z) = \begin{pmatrix} \ddots & & & & & & & & \\ & -\frac{2\pi k}{2\pi|k|} & & & & & & & \\ & & -\frac{2\pi k}{2\pi|k|} & & & & & & \\ & & & \ddots & & & & & \\ & & & & -\frac{2\pi k}{2\pi|k|} & & & & \\ & & & & & -\frac{2\pi k}{2\pi|k|} & & & \\ & & & & & & \ddots & & \end{pmatrix}.$$

The eigenvalues of the Hessian are $\operatorname{sgn} k$ ($k \in \mathbb{Z}$), i.e. ± 1 with infinite multiplicity and 0 with $2n$ multiplicity. There is clearly a "gap" in the spectrum, and the spectrum is also bounded from above and below. Hence, there is exponential decay: Along the flow line converging as $s \to +\infty$, starting at some $s_0 \in \mathbb{R}$, we have

$$d_{W^{\frac{1}{2},2}}(z_s(\cdot), 0) = \|z_s(\cdot)\|_{W^{\frac{1}{2},2}} \leq e^{-(s-s_0)} \|z_{s_0}(\cdot)\|_{W^{\frac{1}{2},2}}.$$

Remark 3.6. Historically, Charles C. Conley and Eduard Zehnder used such fractional Sobolev space setting to study and prove the Arnold conjecture for $\mathbb{T}^{2n}$, c.f. [Abb01, §6.1]. The caveat on the space $W^{\frac{1}{2},2}$ is that there is no Sobolev embedding for the critical exponent $\frac{1}{2}$, so the objects in this space need not to be continuous, i.e. not C^0. Therefore, $W^{\frac{1}{2},2}(\mathbb{S}^1, M)$ is not a Hilbert manifold in general due to lack of smooth structure.

To transfer from flat metric to generic metric, we need to introduce the action-energy inequality. Consider f a Morse–Bott function on a Hilbert manifold (possibly infinite dimension, if finite dimension then it is isomorphic to Euclidean space) with inner product $\langle \cdot, \cdot \rangle$, Under Morse–Bott coordinate, we write:

$$f(x) = \frac{1}{2} \langle x, Ax \rangle$$

where A is the Hessian matrix w.r.t. the inner product which is diagonal with eigenvalues $(a_k)_{k \in \mathbb{Z}}$ as its non-zero entries. The gradient reads $\operatorname{grad} f(x) = Ax$ and

$$\|\operatorname{grad} f(x)\|^2 = \langle Ax, Ax \rangle = \sum_{k \in \mathbb{Z}} a_k^2 x_k^2$$

Assume there is spectrum gap $a := \inf\{|a_k| : a_k \neq 0,\ k \in \mathbb{Z}\} > 0$, then

$$|f(x)| \leq \frac{1}{2}\sum_{k\in\mathbb{Z}} |a_k| x_k^2 = \frac{1}{2a}\sum_{k\in\mathbb{Z}} \underbrace{a|a_k|}_{\leq a_k^2} x_k^2 \leq \frac{1}{2a}\|\operatorname{grad} f(x)\|^2.$$

Set $c := \frac{1}{2a}$, we obtain the ***action-energy inequality***:

$$|f(x)| \leq c\|\operatorname{grad} f(x)\|^2.$$

We see how action-energy inequality for area functional can be derived from isoperimetric inequality: Put $f = \mathscr{A}_0$, $\operatorname{grad}_{L^2} \mathscr{A}_0(z) = i\partial_t z$, action-energy inequality yields:

$$|\mathscr{A}_0(z)| \leq c\|\operatorname{grad}_{L^2} \mathscr{A}_0(z)\|_{L^2}^2 = c\|\partial_t z\|_{L^2}^2.$$

The isoperimetric inequality yields:

$$|\mathscr{A}_0(z)| \leq c' L(z)^2$$

for some constant $c' > 0$, where the length functional

$$L(z) = \int_0^1 |\partial_t z(t)|\, dt = \|\partial_t z\|_{L^1} \overset{\text{Hölder ineq.}}{\leq} \|\partial_t z\|_{L^2}\|1\|_{L^2}.$$

This implies the action-energy inequality.

Lemma 3.11. *Suppose g_1, g_2 are metrics on the smooth manifold, and there exists constant $\kappa > 0$ s.t. $g_1 \leq \kappa^2 g_2$. Then*

$$\|\operatorname{grad}_{g_2} f\|_{g_2} \leq \kappa\|\operatorname{grad}_{g_1} f\|_{g_1}.$$

Remark 3.7. If g_1 and g_2 are conformally equivalent, then $g_1 = \kappa^2 g_2$, and $\operatorname{grad}_{g_1} f = \operatorname{grad}_{\kappa^2 g_2} f = \frac{1}{\kappa^2}\operatorname{grad}_{g_2} f \Rightarrow \|\operatorname{grad}_{g_2} f\|_{g_2} = \kappa\|\operatorname{grad}_{g_1} f\|_{g_1}$.

Lemma 3.12. *Assume f satifies action-energy inequality on some open subset $(U, g|_U)$ in the Riemannian manifold with metric g. Denote $x(s)$ as the rgadient flow line of $\operatorname{grad}_g f$. Then the following hold:*
(i) If $s_- < s_+$ and $x(s) \in U$ with $f(x(s)) > 0\ \forall\, s \in [s_-, s_+]$, then

$$d_g(x(s_-), x(s_+)) \leq \frac{2}{\sqrt{c}}\left(\sqrt{f(x(s_-))} - \sqrt{f(x(s_+))}\right).$$

(ii) Suppose for some $s_0 \in \mathbb{R}$, $x(s) \in U$ with $f(x(s)) \geq 0\ \forall\, s \geq s_0$, then

$$f(x(s)) \leq f(x(s_0))e^{-\frac{s-s_0}{c}}, \qquad \forall\, s \geq s_0.$$

The proof of above lemmas can be found in [FN20, §6.3, §6.4]. The arguments therein work for infinite dimensional scenario and Morse–Bott setup as well. In our concerning case here, the Riemannian manifold is infinite dimensional smooth Hilbert manifold with Riemannian metric induced by the inner product.

The final mission of this section is to show the inequality (17).

Proof of (17). Since M is compact, there exists a finite open cover $U_1, \dots, U_m$ of M s.t. each $\overline{U}_j$ is contained in some Darboux chart. Define open subset in $\mathscr{L}_0 M$

$$\mathscr{U} := \{\gamma \in \mathscr{L}_0 M : \gamma(\mathbb{S}^1) \subseteq U_j \text{ for some } j = 1, \dots, m\}.$$

Step 1: There exists $s_0 \in \mathbb{R}$ such that $w(s,\cdot) \in \mathscr{U}$ for all $s \geq s_0$.

Assume this was not the case, then there exists a sequence $(r_k) \subseteq \mathbb{R}$ with $r_k \to +\infty$ as $k \to \infty$ and $w(r_k,\cdot) \notin \mathscr{U}$ for all $k \in \mathbb{N}$. By Lemma 3.8, there exists $w_\infty \in M$ such that

$$w_k \xrightarrow{C^\infty_{\mathrm{loc}}} w_\infty, \qquad k \to \infty,$$

up to a subsequence. There exists $j = 1, \dots, m$ such that $w_\infty \in U_j$ and $K \in \mathbb{N}$ with $w(r_k, t) \in U_j$ for all $t \in \mathbb{S}^1$ and $k \geq K$. Consequently, we have that $w(r_k,\cdot) \in \mathscr{U}$ for all $k \geq K$. This leads to contradiction to the assumption, and hence finishes Step 1.

Step 2: There exist constants $c > 0$ and $\delta > 0$ such that

$$d_{L^2}(w(s,\cdot), w_\infty) \leq c e^{-\delta s} \qquad \forall s \geq s_0,$$

where w_∞ is the limit of $w(k)$ as $k \to \infty$ up to a subsequence.

The area functional $\mathscr{A}_0$ restricting on $\mathscr{U}$ satisfies the action-energy inequality

$$|\mathscr{A}_0(\gamma)| \leq c_0 \left\|\mathrm{grad}_{J_0} \mathscr{A}_0(\gamma)\right\|^2_{J_0} \qquad \forall \gamma \in \mathscr{U}.$$

Moreover, there exists a constant $\kappa > 0$ s.t. on all $(U_j)_{j=1,\dots,m}$ we have:

$$\frac{1}{\kappa^2} g_J \leq g_{J_0} \leq \kappa^2 g_J.$$

By Lemma 3.11, we conclude

$$|\mathscr{A}_0(\gamma)| \leq c_1 \left\|\mathrm{grad}_{J} \mathscr{A}_0(\gamma)\right\|^2_{J} \qquad \forall \gamma \in \mathscr{U}.$$

For $s \geq s_0$, we choose k_0 s.t. $r_{k_0} > s \geq s_0$. As w is a nonconstant negative gradient flow line of the area functional, we estimate:

$$\begin{aligned}
\mathscr{A}_0(w(s,\cdot)) &> \mathscr{A}_0(w(r_{k_0},\cdot)) \\
&\geq \lim_{k\to\infty} \mathscr{A}_0(w(r_k,\cdot)) \\
&= \lim_{k\to\infty} \mathscr{A}_0(w_k(0,\cdot)) \\
&= \mathscr{A}_0(w_\infty) \\
&= 0.
\end{aligned}$$

Let $s_0 < s_1 < s_2$, with Step 1 and above result, we can now apply Lemma 3.12 and compute

$$\begin{aligned} d_{L^2}(w(s_1,\cdot), w(s_2,\cdot)) &\leq \frac{2}{\sqrt{c_1}} \left(\sqrt{\mathscr{A}_0(w(s_1,\cdot))} - \sqrt{\mathscr{A}_0(w(s_2,\cdot))} \right) \\ &\leq \frac{2}{\sqrt{c_1}} \sqrt{\mathscr{A}_0(w(s_1,\cdot))} \\ &\leq \frac{2}{\sqrt{c_1}} \sqrt{\mathscr{A}_0(w(s_0,\cdot))} e^{\frac{1}{2c_1}(s_0 - s_1)} \end{aligned}$$

Choose k_1 s.t. $r_{k_1} \geq s_1$. Then for all $k \geq k_1$, we have that

$$d_{L^2}(w(s_1,\cdot), w(r_k,\cdot)) \leq c e^{-\delta s_1}$$

with $c := \frac{2}{\sqrt{c_1}} \sqrt{\mathscr{A}_0(w(s_0,\cdot))} e^{\frac{s_0}{2c_1}}$, $\delta := \frac{1}{2c_1}$, and thus

$$d_{L^2}(w(s_1,\cdot), w_\infty) = \lim_{j \to \infty} d_{L^2}(w(s_1,\cdot), w_{r_{k_j}}(0,\cdot)) \leq c e^{-\delta s_1}.$$

This completes Step 2 and hence the proof. □

4 Hamiltonian Floer Homology

4.1 Ungraded Floer Homology

Let (M,ω) be a symplectically aspherical compact symplectic manifold and $H \in C^\infty(M\times\mathbb{S}^1)$ nondegenerate. Define the finite-dimensional $\mathbb{Z}_2$-vector space, called the ***Floer chain group***, by

$$\mathrm{CF}(H) := \mathscr{P}_0(H) \otimes \mathbb{Z}_2.$$

Note that the set $\mathscr{P}_0(H) = \operatorname{Crit}\mathscr{A}_H$ is now equipped with formal sum and therefore considered as a free $\mathbb{Z}$-module. For a smooth family $J = (J_t)_{t\in\mathbb{S}^1}$ of ω-compatible almost complex structures and $\gamma^\pm \in \mathscr{P}_0(H)$ define the ***moduli space of parametrised negative gradient flow lines from γ^- to γ^+*** by

$$\widetilde{\mathcal{M}}_J(\gamma^-,\gamma^+) := \{u \in C^\infty(\mathbb{R}\times\mathbb{S}^1, M) : u \text{ solves (7) and } \lim_{s\to\pm\infty} u(s,\cdot) = \gamma^\pm\}.$$

If $\gamma^+ \neq \gamma^-$, we can consider the ***moduli space of unparametrised negative gradient flow lines from γ^- to γ^+***

$$\mathcal{M}_J(\gamma^-,\gamma^+) := \widetilde{\mathcal{M}}_J(\gamma^-,\gamma^+)/\mathbb{R},$$

where the free $\mathbb{R}$-action on the moduli space $\widetilde{\mathcal{M}}_J(\gamma^-,\gamma^+)$ is given by timeshift, that is,

$$(r\cdot u)(s,t) := u(s+r,t) \qquad \forall (s,t)\in\mathbb{R}\times\mathbb{S}^1.$$

For $u \in \widetilde{\mathcal{M}}_J(\gamma^-,\gamma^+)$, one can consider the Fredholm operator D_u given by linearising the Floer equation (7) at u. If J is choosen generically such that D_u is surjective for all negative gradient flow lines u, then $\widetilde{\mathcal{M}}_J(\gamma^-,\gamma^+)$ is a smooth manifold of local dimension

$$\dim_u \widetilde{\mathcal{M}}_J(\gamma^-,\gamma^+) = \operatorname{ind} D_u.$$

Define a linear map

$$\partial\colon \mathrm{CF}(H)\to\mathrm{CF}(H), \qquad \partial\gamma^- := \sum_{\gamma^+\in\mathscr{P}_0(H)} n(\gamma^-,\gamma^+)\gamma^+, \tag{18}$$

where

$$n(\gamma^-,\gamma^+) := \#_2\mathcal{M}^0_J(\gamma^-,\gamma^+) \in \mathbb{Z}_2,$$

with $\mathcal{M}^0_J(\gamma^-,\gamma^+)$ denoting the zero-dimensional component of $\mathcal{M}_J(\gamma^-,\gamma^+)$. In order to show that (18) is well-defined we observe that $\mathcal{M}^0_J(\gamma^-,\gamma^+)$ is a finite set. Indeed, $\mathcal{M}^0_J(\gamma^-,\gamma^+)$ is compact by [MS12, Theorem 4.1.1] as by Theorem 3.4 the derivatives of the negative gradient flow lines are uniformly bounded and there do not occur broken negative gradient flow lines for index reasons. In fact, ∂ is a chain map, that is, $\partial^2 = 0$. This can be seen as follows. Observe that $\mathcal{M}^1_J(\gamma^-,\gamma^+)$ cannot be compact as in the limit we can have once broken negative gradient flow lines. However, one can compactify the one-dimensional

moduli space of unparametrised negative gradient flow lines to a compact one-dimensional manifold with boundary $\overline{\mathcal{M}_J^1(\gamma^-,\gamma^+)}$ whose boundary consits of broken flow lines

$$\partial\overline{\mathcal{M}_J^1(\gamma^-,\gamma^+)} \cong \coprod_{\gamma\in\mathscr{P}_0(H)} \mathcal{M}_J^0(\gamma^-,\gamma)\times\mathcal{M}_J^0(\gamma,\gamma^+)$$

via gluing. Thus we compute

$$\begin{aligned}\partial^2\gamma^- &= \partial\left(\sum_{\gamma\in\mathscr{P}_0(H)} n(\gamma^-,\gamma)\gamma\right)\\ &= \sum_{\gamma^+\in\mathscr{P}_0(H)}\sum_{\gamma\in\mathscr{P}_0(H)} n(\gamma^-,\gamma)n(\gamma,\gamma^+)\gamma^+\\ &= \sum_{\gamma^+\in\mathscr{P}_0(H)} \#_2\partial\overline{\mathcal{M}_J^1(\gamma^-,\gamma^+)}\gamma^+\\ &= 0,\end{aligned}$$

as by the classification of compact one-dimensional smooth manifolds the only such manifolds are disjoint unions of circles and closed intervals. Thus we can define the ***Hamiltonian Floer homology of (M,ω) and H*** to be the $\mathbb{Z}_2$-vector space given by

$$\mathrm{HF}(H) := \frac{\ker\partial}{\operatorname{im}\partial}.$$

An analogous invariance argument via continuation homomorphisms as is Morse homology shows that the dimension of $\mathrm{HF}(H)$ is independent of the choice of the generic family J and nondegenerate Hamiltonian function H. In order to explicitly compute the Floer homology on a symplectically aspherical compact symplectic manifold (M,ω), we use a slight generalisation of this construction. In fact, one can construct a Morse homology for every $H\in C^\infty(M\times\mathbb{S}^1)$ such that the associated symplectic action functional $\mathscr{A}_H$ is Morse–Bott. In this scenario, we choose an additional Morse function on $\operatorname{Crit}\mathscr{A}_H\subseteq M$ and count negative gradient flow lines with cascades. We are now in position to prove the homological Arnold conjecture.

Theorem 4.1 (Homological Arnold Conjecture in the Symplectically Aspherical Case). *Let (M,ω) be a symplectically aspherical compact symplectic manifold and $H\in C^\infty(M\times\mathbb{S}^1)$ nondegenerate. Then*

$$\#\mathscr{P}_0(H) \geq \sum_{k=0}^{\dim M} \dim \mathrm{H}_k(M;\mathbb{Z}_2), \tag{19}$$

where $\mathscr{P}(H)$ denotes the set of all 1-periodic orbits of H.

Proof. The inequality (19) follows from the chain of natural isomorphisms

$$\mathrm{HF}(H)=\mathrm{HM}(\mathscr{A}_H)=\mathrm{HM}(\mathscr{A}_H,0)\cong\mathrm{HM}(\mathscr{A}_0,h)=\mathrm{HM}(h)=\bigoplus_{k\in\mathbb{Z}}\mathrm{HM}_k(h)\cong\bigoplus_{k\in\mathbb{Z}}\mathrm{H}_k(M).$$

□

4.2 Graded Floer Homology

In the end, we discuss grading for Floer homology under suitable assumptions. In general, there is no $\mathbb{Z}$-grading as in Morse homology. However, we always a $\mathbb{Z}_2$-grading which is suitable for some applications. Let $\gamma \in \operatorname{Crit} \mathscr{A}_H$ and choose a filling disk $\bar{\gamma} \in C^\infty(\mathbb{D}, M^{2n})$ of γ. Moreover, choose a symplectic trivialisation for the pullback tangent bundle $\Phi\colon \mathbb{D} \times \mathbb{R}^{2n} \to \bar{\gamma}^* TM$. This is possible by [MS17, Proposition 2.6.7]. Then we can associated to the pair $(\gamma, \bar{\gamma})$ an integer, called the Conley–Zehnder index of γ (c.f. [FK18, Definition 10.4.1]), by

$$\mu_{\mathrm{CZ}}(\gamma, \bar{\gamma}) := \mu_{\mathrm{CZ}}(\Psi),$$

where

$$\Psi\colon [0,1] \to \mathrm{Sp}(n), \qquad \Psi(t) := \Phi_{e^{2\pi i t}}^{-1} \circ D\phi_t^{X_H} \circ \Phi_1.$$

If we have another filling disk $\bar{\gamma}'$ of γ, then

$$\mu_{\mathrm{CZ}}(\gamma, \bar{\gamma}) - \mu_{\mathrm{CZ}}(\gamma, \bar{\gamma}') = 2c_1([\bar{\gamma}' \# (-\bar{\gamma})]),$$

where $[\bar{\gamma}' \# (-\bar{\gamma})] \in \pi_2(M)$. So we have a well-defined $\mathbb{Z}_2$-grading in general or a well-defined $\mathbb{Z}$-grading if $c_1|_{\pi_2(M)} = 0$. For example, this happens for the torus $\mathbb{T}^{2n}$ as $\pi_2(\mathbb{T}^{2n}) = 0$.

Alternatively, there is a natural $\mathbb{Z}_2$-grading which can be defined via monodromy matrices. Let (M, ω) be a symplectic manifold and $H \in C^\infty(M \times \mathbb{S}^1)$. Consider the linearisation

$$D\phi_1^H|_{\gamma(0)}\colon T_{\gamma(0)}M \to T_{\gamma(0)}M$$

of the flow of the Hamiltonian vector field X_H for some 1-periodic orbit γ or equivalently, a fixed point $x = \gamma(0)$ of the Hamiltonian symplectomorphism $\varphi_H := \phi_1^H$. Assume that (M, ω) is symplectically aspherical and H nondegenerate. Recall the nondegeneracy condition yields

$$\det(D\varphi_H|_{\gamma(0)} - \mathrm{id}_{T_{\gamma(0)}M}) \neq 0 \qquad \forall\, \gamma \in \mathscr{P}_0(H).$$

Thus, the set $\mathscr{P}_0(H)$ can be decomposed into two disjoint substes:

$$\mathscr{P}_0^+(H) := \{\gamma \in \mathscr{P}_0(H) : \det(D\varphi_H|_{\gamma(0)} - \mathrm{id}_{T_{\gamma(0)}M}) > 0\},$$
$$\mathscr{P}_0^-(H) := \{\gamma \in \mathscr{P}_0(H) : \det(D\varphi_H|_{\gamma(0)} - \mathrm{id}_{T_{\gamma(0)}M}) < 0\}.$$

Then we define

$$\mathrm{CF}^\pm(H) := \mathscr{P}_0^\pm(H) \otimes \mathbb{Z}_2 \qquad \text{and} \qquad \mathrm{CF}(H) := \mathrm{CF}^-(H) \oplus \mathrm{CF}^+(H).$$

Moreover, the usual boundary operator of Floer homology gives rise to restrictions

$$\partial^\pm\colon \mathrm{CF}^\pm(H) \to \mathrm{CF}^\mp(H).$$

Thus we get two Floer homology groups

$$\mathrm{HF}^\pm(H) := \frac{\ker \partial^\pm}{\operatorname{im} \partial^\mp}$$

and an improved version of the homological Arnold conjecture

$$\#\mathscr{P}_0^+(H) \geq \sum_{k \text{ even}} \dim \mathrm{H}_k(M;\mathbb{Z}_2) \quad \text{and} \quad \#\mathscr{P}_0^-(H) \geq \sum_{k \text{ odd}} \dim \mathrm{H}_k(M;\mathbb{Z}_2)$$

5 Rabinowitz Floer Homology

A Morse–Bott Homology

Morse–Bott homology is a generalisation of Morse homology to functions with degenerate critical points which are still sufficiently well behaved. See [CF09, Appendix A] for a short introduction to Morse–Bott homology and [Fra04, Appendix A] for a more extensive treatment. Morse–Bott functions often occur in the presence of symmetries. Indeed, let G be a Lie group acting smoothly on a manifold M. If a function $f \in C^\infty(M)$ is G-invariant, that is, $f(gx) = f(x)$ holds for all $g \in G$ and $x \in M$, then $\operatorname{Crit} f$ is also G-invariant. In particular, f is not a Morse function in general. However, the presence of symmetry usually simplyfies the explicit computation of the Morse homology.

Definition A.1 (Morse–Bott Function). *Let M be a smooth manifold. A **Morse–Bott function on M** is defined to be a function $f \in C^\infty(M)$ such that*

(i) $\operatorname{Crit} f \subseteq M$ *is an embedded submanifold (consisting of disjoint connected components).*

(ii) $T_x \operatorname{Crit} f = \ker \operatorname{Hess} f|_x$ *for all* $x \in \operatorname{Crit} f$.

Remark A.1. Assumption (ii) on a Morse–Bott function is crucial for proving the Morse–Bott Lemma [BH04, Lemma 3.51], an analogue of the Morse Lemma. The Morse–Bott Lemma is a key technical ingredient for proving exponential decay of gradient flow lines.

Let M be a compact smooth manifold and $f \in C^\infty(M)$ a Morse–Bott function. Choose an additional Morse function h on $\operatorname{Crit} f$ such that h satisfies the Morse–Smale condition. Define a $\mathbb{Z}$-graded chain complex consisting of graded vector spaces

$$\mathrm{CM}_k(f,h) := \left\{ \sum_{\substack{x \in \operatorname{Crit} h \\ \operatorname{ind}^m(x) = k}} a_x x : a_x \in \mathbb{Z}_2 \right\}, \qquad k \in \mathbb{Z}$$

with the grading index ind^m defined by

$$\operatorname{ind}^m(x) := \operatorname{ind}_f(x) + \operatorname{ind}_h(x) \qquad \forall x \in \operatorname{Crit} h,$$

where ind denotes the ordinary Morse index. The boundary operator is defined by

$$\partial_k \colon \mathrm{CM}_k(f,h) \to \mathrm{CM}_{k-1}(f,h), \qquad \partial_k x^- := \sum_{\substack{x^+ \in \operatorname{Crit} h \\ \operatorname{ind}^m(x^+) = k-1}} n(x^-, x^+) x^+,$$

where

$$n(x,y) := \#_2 \mathcal{M}(x^-, x^+) \in \mathbb{Z}_2$$

with $\mathcal{M}(x^-, x^+)$ denoting the moduli space of unparametrised gradient flow lines with cascades from x^- to x^+. A cascade starts with a gradient flow line of the Morse function h on $\operatorname{Crit} f$ and meets in finite time the negatively asymptotic end of a negative gradient flow line of the Morse–Bott function f. This is called the first cascade. At the positive asymptotic end, this Morse–Bott gradient flow line meets again a gradient flow line of h. After finite time, a second cascade might appear, but after finitely many cascades, the gradient flow of h converges asymptotically to a critical point of h. By [Fra04, Theorem A.12], $\mathcal{M}(x^-, x^+)$ is generically a smooth manifold of dimension

$$\dim \mathcal{M}(x^-, x^+) = \operatorname{ind}^m(x^-) - \operatorname{ind}^m(x^+) - 1,$$

and thus $n(x^-, x^+)$ is well-defined. This dimensional formula comes from the Fredholm index of the linearised negative gradient flow equation. Define now the Morse–Bott homology of M by

$$\operatorname{HM}_k(f, h) := \frac{\ker \partial_k}{\operatorname{im} \partial_{k+1}} \qquad \forall k \in \mathbb{Z}.$$

The homology group $\operatorname{HM}(f, h)$ is independent of any auxiliary data up to natural isomorphisms. In particular, as every Morse function is a Morse–Bott function, Morse–Bott homology is canonically isomorphic to the ordinary Morse homology of M, and thus in particular to the singular homology of M with coefficients in $\mathbb{Z}_2$, that is

$$\operatorname{HM}_*(f, h) \cong \operatorname{H}_*(M; \mathbb{Z}_2).$$

Alternatively, one obtains a $\mathbb{Z}$-grading of Morse–Bott homology by considering the signature index

$$\operatorname{ind}^\sigma(x) := -\frac{1}{2} \operatorname{sgn} \operatorname{Hess} f(x) - \frac{1}{2} \operatorname{sgn} \operatorname{Hess} h(x) \qquad \forall x \in \operatorname{Crit} h.$$

The Morse index and the signature index are related by the global shift

$$\operatorname{ind}^\sigma = \operatorname{ind}^m - \frac{1}{2} \dim M.$$

B Polyfolds

In order to deal with the generic transversality of the moduli space of negative gradient flow lines of the symplectic action functional it is convenient to introduce the language of polyfolds. One can deal with such questions also classically, but the abstract approach via polyfolds is much more general. Given $\gamma^{\pm} \in \operatorname{Crit} \mathcal{A}_H$, $\gamma^- \neq \gamma^+$, one considers the moduli space

$$\widetilde{\mathcal{M}}_J(\gamma^-, \gamma^+)$$

of parametrised negative gradient flow lines of $\mathcal{A}_H$ for some compatible almost complex structure J. The natural free $\mathbb{R}$-action given by time-shift yields the moduli space

$$\mathcal{M}_J(\gamma^-, \gamma^+) := \widetilde{\mathcal{M}}_J(\gamma^-, \gamma^+)/\mathbb{R}$$

of unparametrised negative gradient flow lines. In order to show that those two moduli spaces are smooth manifolds, we want to write them as the zero level set of a section of a suitable Hilbert bundle. In ordinary Morse homology, this would be

$$s \colon \mathcal{H} \to \mathcal{E}, \qquad s(x) := \dot{x} + \operatorname{grad} f(x),$$

where $\mathcal{H}$ is the Hilbert manifold of all $W^{1,2}$-paths from x^- to x^+, $x^{\pm} \in \operatorname{Crit} f$, for some Morse function f, and $\mathcal{E}_x := L^2(\mathbb{R}, x^*TM)$. Then

$$\widetilde{\mathcal{M}}(x^-, x^+) = s^{-1}(0).$$

Generically, this moduli space is a smooth manifold by using standard transversality techniques due to Taubes. However, this results in a problem, as the time shift map

$$\mathbb{R} \times \mathcal{H} \to \mathcal{H}, \qquad (r, x) \mapsto x(\cdot + r)$$

is not smooth. Indeed, in general we only have that $\partial_r r_* x = \dot{x} \in L^2$. Thus it is convenient to introduce a new notion of smoothness, tailored to make the time shift map smooth. We follow [FW21].

Definition B.1 (Scale Structure). *A **scale structure** on a Banach space E is a decreasing sequence*

$$E =: E_0 \supseteq E_1 \supseteq E_2 \supseteq \dots$$

*of Banach spaces such that the inclusion $E_{k+1} \hookrightarrow E_k$ is compact for every $k \in \mathbb{N}_0$ and such that $E_\infty := \cap_{k=0}^{\infty} E_k$ is dense in every E_k. A Banach space E together with a scale structure (E_k) is called a **scale Banach space**.*

Example B.1 (Shifted Scale Banach Space). Let $(E, (E_k))$ be a scale Banach space and $m \in \mathbb{N}_0$. Then $(E^k, (E_k^m))$ is a scale Banach space where $E_k^m := E_{m+k}$ for all $k \in \mathbb{N}_0$.

Example B.2 (Scale Direct Sum). Let $(E, (E_k))$ and $(F, (F_k))$ be scale Banach spaces. Then $(E \oplus F, (E_k \oplus F_k))$ is also a scale Banach space.

The following example underlies Morse and Floer theory.

Example B.3 (Weighted Sobolev Spaces). Fix a monotone cutoff function $\beta \in C^\infty(\mathbb{R}, [-1,1])$ with

$$\beta(s) = \begin{cases} 1 & s \geq 1, \\ -1 & s \leq -1, \end{cases}$$

and $\delta > 0$. Define

$$\gamma_\delta \colon \mathbb{R} \to \mathbb{R}, \qquad \gamma_\delta(s) := e^{\delta\beta(s)s}.$$

For $p \in (1, +\infty)$ and $k \in \mathbb{N}_0$ define the ***Sobolev spaces with weight δ*** by

$$W^{k,p}_\delta(\mathbb{R}, \mathbb{R}^n) := \{u \in W^{k,p}(\mathbb{R}, \mathbb{R}^n) : \gamma_\delta u \in W^{k,p}(\mathbb{R}, \mathbb{R}^n)\}.$$

Choose a strictly increasing sequence (δ_k) with $\delta_0 = 0$. Then $(E, (E_k))$ is a scale Banach space with

$$E_k := W^{k,p}_{\delta_k}(\mathbb{R}, \mathbb{R}^n) \qquad \forall k \in \mathbb{N}_0.$$

Definition B.2 (Scale Continuity). *Let $(E, (E_k))$ and $(F, (F_k))$ be scale Banach spaces. A map $f \colon E \to F$ is called **scale continuous**, iff $f(E_k) \subseteq F_k$ for all $k \in \mathbb{N}_0$ and the restriction $f|_{E_k} \colon E_k \to F_k$ is continuous.*

Definition B.3 (Tangent Bundle). *For a scale Banach space $(E, (E_k))$ define its **tangent bundle** by $TE := E^1 \oplus E^0$.*

Definition B.4 (Scale Differentiability). *A scale continuous map $f \colon E \to F$ between scale Banach spaces $(E, (E_k))$ and $(F, (F_k))$ is called **scale differentiable**, iff for every $x \in E_1$ there exists a bounded linear operator*

$$Df(x) \colon E_0 \to F_0,$$

*called the **scale differential of f**, such that the restriction $f|_{E_1} \colon E_1 \to F_0$ is Fréchet differentiable with derivative $Df|_{E_1}$ and the **tangent map***

$$Tf \colon TE \to TF, \qquad Tf(x,h) := (f(x), Df(x)h)$$

is scale continuous.

Using the iterated notion of scale differentiability one can define higher scale regularity. We say that a scale differentiable map is ***scale smooth***, iff its tangent map is infinitely scale differentiable.

Example B.4 (Weighted Sobolev Spaces). Let E_k be the scale of weighted Sobolev spaces introduced in Example B.3 on the scale Banach space $E = L^p(\mathbb{R}, \mathbb{R}^n)$. Then the shift map

$$\mathbb{R} \oplus E \to E, \qquad (r, x) := x(\cdot + r)$$

is scale smooth.

One important property of scale differentiability is that the chain rule is still valid in this general setting.

Proposition B.1 (Chain Rule). *Let $f\colon E \to F$ and $g\colon F \to G$ be scale differentiable maps for scale Banach spaces E, F and G. Then the composition $g \circ f$ is scale differentiable with tangent map*

$$T(g \circ f) = Tg \circ Tf\colon TE \to TG.$$

The proof of the chain rule heavily relies on the compactness of the embeddings of the scales. Using the chain rule one is able to define the notion of scale manifolds and scale differential geometry in analogy to the finite-dimensional case. Thus with the theory developed so far, one can make sense of the smooth moduli space of unparametrised negative gradient flow lines. However, for broken negative gradient flow lines one needs an even more general notion.

Definition B.5 (Retraction). *Let X b a topological space. A **retraction** is a continuous map $R\colon X \to X$ such that $R^2 = R$.*

Remark B.1. By a result of Cartan, if X is a smooth Hilbert manifold, then $\mathrm{Fix}(R)$ is a smooth submanifold for every smooth retraction $R\colon X \to X$.

It is in general not true that the fixed point set of a scale smooth retraction of a scale manifold is a scale submanifold. This lead Hofer–Wysocki–Zehnder to the generalised notion of an ***M-polyfold***, where the "M" stands for "manifold flavoured". Heuristically, an M-polyfold is locally the fixed point set of a scale smooth retraction of a scale smooth manifold. The main aspect for us is that Fredholm theory still is valid in M-polyfolds in some sense. For an extensive treatment see [HWZ21]. Without details, we expalin now the M-polyfold Fredholm setup of Morse–Bott homology, see Theorem B.1.

We assume the following setup. Let (M, m) be a compact Riemannian manifold and $f \in C^\infty(M)$ a Morse–Bott function. Choose a Morse–Smale pair (h, m_0) on $\mathrm{Crit}\, f$, that is, $h \in C^\infty(\mathrm{Crit}\, f)$ is a Morse function and $(\mathrm{Crit}\, f, m_0)$ is a Riemannian manifold such that the stable and unstable manifolds of h intersect transversally. Pick two connected critical components $C_\pm \subseteq \mathrm{Crit}\, f$. For

$$0 < \delta < \min\{|\lambda| : \lambda \in \sigma(\mathrm{Hess}_x f)) \setminus \{0\}, x \in \mathrm{Crit}\, f\}$$

and a positive strictly increasing sequence $0 < \delta_0 < \delta_1 < \cdots < \delta$, consider the Banach manifold $\tilde{\mathcal{E}}_k(C_-, C_+)$ consisting of all maps $u \in H^{k+2}_{\delta_k}(\mathbb{R}, M)$ converging exponentially to $C_\pm$, that is, there exist $x_\pm \in C_\pm$ as well as

$$\xi_- \in H^{k+2}_{\delta_k}((-\infty, -T], T_{x_-}M) \quad \text{and} \quad \xi_+ \in H^{k+2}_{\delta_k}([T, +\infty), T_{x_+}M)$$

for some $T \in \mathbb{R}$ with

$$u(s) = \exp_{x_\pm}(\xi_\pm(s)) \qquad \forall \pm s \geq T.$$

See Figure 8. Local charts on $\tilde{\mathcal{E}}_k(C_-, C_+)$ are constructed by exponential neighbourhoods around smooth paths [Sch93, Appendix A]. Using those exponential

neighbourhoods, one can construct scale smooth charts, making $\tilde{\mathcal{E}}(C_-, C_+) = \tilde{\mathcal{E}}_0(C_-, C_+)$ into a scale smooth manifold. The construction is a much more involved argument similar to [AW13] in the case of fixed asymptotics. Moreover, there are two natural scale smooth evaluation maps

$$\mathrm{ev}_\pm \colon \tilde{\mathcal{E}}(C_-, C_+) \to \operatorname{Crit} f, \qquad \mathrm{ev}_\pm(u) = x_\pm.$$

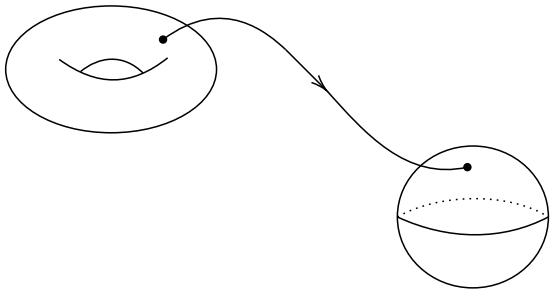


Figure 8: An asymptotically exponential Sobolev path connecting two critical components of a Morse–Bott function.

Fix $x_\pm \in C_\pm \cap \operatorname{Crit} h$ and define the stable and unstable manifolds by

$$W^\pm(x_\pm) := \left\{ x \in \operatorname{Crit} f : \lim_{s \to \pm\infty} \phi_s(x) = x_\pm \right\},$$

where $\phi \colon \mathbb{R} \times \operatorname{Crit} f \to \operatorname{Crit} f$ denotes the negative gradient flow of h with respect to the Morse–Smale metric m_0. Define

$$\tilde{\mathcal{E}}^1_k(x_-, x_+) := \left\{ u \in \tilde{\mathcal{E}}_k(C_-, C_+) : \mathrm{ev}_\pm(u) \in W^\pm(x_\pm) \right\}$$

for all $k \in \mathbb{N}_0$. See Figure 9. Again, $\tilde{\mathcal{E}}^1(x_-, x_+) := \tilde{\mathcal{E}}^1_0(x_-, x_+)$ is a scale smooth manifold.

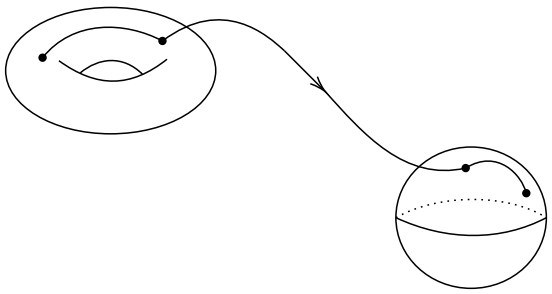


Figure 9: An asymptotically exponential Sobolev path connecting two critical components of a Morse–Bott function.

There exists a canonical scale smooth bundle

$$\tilde{\mathcal{F}}(x_-, x_+) \to \tilde{\mathcal{E}}(x_-, x_+),$$

where the fibre over $u \in \tilde{\mathcal{E}}_k(C_-, C_+)$ is given by

$$\tilde{\mathcal{F}}_k(x_-, x_+)|_u = H^{k+1}_{\delta_k}(\mathbb{R}, u^*TM)$$

for all $k \in \mathbb{N}_0$. Then

$$\tilde{\partial}\colon \tilde{\mathcal{E}}(x_-, x_+) \to \tilde{\mathcal{F}}(x_-, x_+), \qquad \tilde{\partial}(u) := \dot{u} + \operatorname{grad}_m f(u)$$

is a scale Fredholm operator. If $x_- \neq x_+$, then there is a free scale smooth reparametrisation action

$$\mathbb{R} \oplus \tilde{\mathcal{E}}(x_-, x_+) \to \tilde{\mathcal{E}}(x_-, x_+), \qquad (r, u) \mapsto u(\cdot + r),$$

as in Example B.4, giving rise to a scale smooth bundle

$$\tilde{\mathcal{F}}(x_-, x_+)/\mathbb{R} \to \tilde{\mathcal{E}}(x_-, x_+)/\mathbb{R}$$

to which $\tilde{\partial}$ descends to a scale Fredholm section ∂. Likewise we get smooth scale bundles of unbroken paths from x^- to x^+ with k cascades by

$$\mathcal{E}^k(x^-, x^+) := \widetilde{\mathcal{E}}^k(x^-, x^+)/\mathbb{R}^k \quad \text{and} \quad \mathcal{F}^k(x^-, x^+) := \widetilde{\mathcal{F}}^k(x^-, x^+)/\mathbb{R}^k,$$

where the additive group $\mathbb{R}^k$ acts by timeshift on the cascades. The scale Fredholm section $\widetilde{\partial}$ descends to define a scale Fredholm section

$$\partial\colon \mathcal{E}^k(x^-, x^+) \to \mathcal{F}^k(x^-, x^+).$$

Now we consider broken paths. Let $x^0, \dots, x^l \in \operatorname{Crit} h$ with

$$f(x^0) > f(x^1) > \cdots > f(x^{l-1}) > f(x^l).$$

Then we can conisder the space of broken paths from x^0 to x^l defined by

$$\mathcal{E}^k(x^0, \dots, x^l) := \mathcal{E}^k(x^0, x^1) \times \cdots \times \mathcal{E}^k(x^{l-1}, x^l).$$

Again, we have a scale Fredholm section

$$\partial \times \cdots \times \partial\colon \mathcal{E}^k(x^0, \dots, x^l) \to \mathcal{F}^k(x^0, \dots, x^l),$$

where

$$\mathcal{F}^k(x^0, \dots, x^l) := \mathcal{F}^k(x^0, x^1) \times \cdots \times \mathcal{F}^k(x^{l-1}, x^l).$$

Therefore, for $x^\pm \in \operatorname{Crit} h$ with $f(x^-) > f(x^+)$ we can define the space of broken paths from x^- to x^+ by

$$X(x^-, x^+) := \coprod_{k \in \mathbb{N}} \coprod_{l \in \mathbb{N}} \coprod_{f(x^-) > f(x^1) > \cdots > f(x^l) > f(x^+)} \mathcal{E}^k(x^-, x^1, \dots, x^l, x^+)$$

Similarly, one defines

$$Y(x^-,x^+) := \coprod_{k\in\mathbb{N}} \coprod_{l\in\mathbb{N}} \coprod_{f(x^-)>f(x^1)>\cdots>f(x^l)>f(x^+)} \mathcal{F}^k(x^-,x^1,\dots,x^l,x^+).$$

One can show that $X(x^-,x^+)$ admits a natural structure as an M-polyfold and

$$Y(x^-,x^+) \to X(x^-,x^+)$$

is a bundle. Moreover, the scale Fredholm section ∂ induces a scale Fredholm section

$$f\colon X(x^-,x^+) \to Y(x^-,x^+)$$

of index

$$\operatorname{ind} f = \operatorname{ind}^m(x^-) - \operatorname{ind}^m(x^+) - 1 = \operatorname{ind}^\sigma(x^-) - \operatorname{ind}^\sigma(x^+) - 1,$$

where

$$\operatorname{ind}^\sigma(x^\pm) := -\frac{1}{2}\operatorname{sgn}\operatorname{Hess} f(x^\pm) - \frac{1}{2}\operatorname{sgn}\operatorname{Hess} h(x^\pm)$$

denotes the signature index, and such that $f^{-1}(0)$ is compact. Finally, if we define

$$X := \coprod_{f(x^-)>f(x^+)} X(x^-,x^+) \qquad \text{and} \qquad Y := \coprod_{f(x^-)>f(x^+)} Y(x^-,x^+),$$

one can show that there exists a scale section $s\colon X \to Y$ such that $f+s$ is a scale Fredholm section. Moreover, we have the following abstract perturbation result for Morse–Bott homology.

Theorem B.1 (M-Polyfold Fredholm Setup for Morse–Bott Homology). *Let (M,m) be a compact Riemannian manifold without boundary and $f \in C^\infty(M)$ a Morse–Bott function. Choose an additional Morse function $h \in C^\infty(\operatorname{Crit} f)$ and an additional Riemannian metric m_0 on $\operatorname{Crit} f$. For all $x^\pm \in \operatorname{Crit} h$, the moduli space*

$$\mathcal{M}(x^-,x^+) := \{u \in X(x^-,x^+) : f(u) + s(u) = 0\}$$

is a smooth compact manifold with corners of dimension

$$\dim \mathcal{M}(x^-,x^+) = \operatorname{ind}^m(x^-) - \operatorname{ind}^m(x^+) - 1 = \operatorname{ind}^\sigma(x^-) - \operatorname{ind}^\sigma(x^+) - 1.$$

Moreover, there is a canoncial diffeomorphism

$$\partial\mathcal{M}(x^-,x^+) = \mathcal{M}(x^-,x^+) \cap \partial X \cong \coprod_{x\in\operatorname{Crit} h} \mathcal{M}(x^-,x) \times \mathcal{M}(x,x^+).$$

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